

From Fugitive Cash to Non-Fugitive Provisioning: Redistribution Feasibility under International Settlement Constraints

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Abstract

Automation is reducing labour demand across occupations and countries simultaneously — cognitive work through large language models, physical work through autonomous machinery and robotics — closing the traditional channel through which displaced workers move to surviving sectors and making large-scale redistribution a structural necessity. Existing models of universal basic income and cash transfers treat feasibility as a question of domestic fiscal capacity, leaving the international monetary constraint unexamined. This paper constructs a twenty-five-bloc closed-world computational model in which every import is another bloc’s export and every capital outflow is another’s inflow. Households allocate income across food, general consumption, and housing through a Stone–Geary demand system. Four redistributive policy regimes — universal cash, targeted cash, food vouchers, and in-kind housing via community land trusts — are calibrated to deliver identical welfare at introduction. A foreign-exchange settlement constraint binds non-reserve blocs: imports cannot exceed the sum of export earnings, reserves, and bor-

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rowing capacity. The model produces five findings. First, welfare-equivalent regimes generate sharply different macrofinancial footprints because their demand composition and import content differ structurally. Second, in blocs with low creditworthiness and low food self-sufficiency, universal cash transfers make recipients worse off than no transfer at all. Third, the feasible set of regimes varies by position on a creditworthiness–self-sufficiency plane, producing an institutional feasibility phase diagram. Fourth, automation amplifies this selection pressure, shrinking the feasible set and separating the regimes. Fifth, cost-of-living adjustment creates a world-price externality: each bloc’s individually rational indexation collectively undermines purchasing power. The international monetary system selects not only how much redistribution a country can afford but which institutional forms can survive. *Keywords:* redistribution, settlement constraint, community land trust, automation, monetary hierarchy, transfer modality

1. Introduction

Automation and artificial intelligence are reducing the demand for human labour across occupations and countries simultaneously. Office work is being restructured by large language models whose multilingual capability allows industry-specific software built under the strict regulatory regimes of high-income countries to diffuse rapidly into the high-revenue segments of developing economies with minimal localisation cost. Manufacturing is being restructured by autonomous machinery, logistics by self-driving vehicles, and physical services by humanoid robots and drones. Previous waves of automation displaced one broad occupational class at a time, and the dis-

placed workers moved to the surviving class; this time, cognitive and physical automation are advancing concurrently, closing the traditional adjustment channel. These are not prospective disruptions but extensions of changes already underway. Developing countries that have not yet completed industrialisation face the additional pressure of premature deindustrialisation: manufacturing employment peaks at lower shares and lower income levels than it did for earlier developers, as labour-saving technology and globalisation close the traditional pathway from agriculture to industry (Rodrik, 2016). When these forces operate concurrently across the world economy, the result is a simultaneous contraction in labour income that no single country caused and no single country can reverse through domestic policy alone.

When labour income falls, people must be provisioned through some other channel. The question is not whether redistribution will be needed — it will — but whether every country can sustain it. A government that can neither restore employment nor deliver transfers loses credibility. Economic collapse and political instability do not follow a fixed sequence: default may precede civil disorder, or civil disorder may destroy the economy that sustains the state. In either case, the consequences do not remain within national borders. They take the form of armed conflict, coercive migration, or refugee flows that impose costs on neighbouring and distant states alike. Preventing this outcome at multiple points across the world economy simultaneously is the problem that motivates this research.

The dominant framework for analysing redistribution treats it as a question of domestic fiscal capacity. Models of universal basic income (UBI) and other cash transfers typically operate within a closed economy or a sin-

gle small open economy facing an exogenous rest of the world. They ask whether tax revenue can cover the transfer, what happens to labour supply, and how the income distribution shifts (Darulich and Fernández, 2024; Luduvic, 2024; Banerjee, Niehaus, and Suri, 2019; Coady and Prady, 2018). Even heterodox approaches that recognise the external constraint — the post-Keynesian literature on monetary sovereignty and currency hierarchy — have directed their analysis at the job guarantee rather than at cash or in-kind transfers (Bonizzi, Kaltenbrunner, and Michell, 2019; Prates, 2020; Vernengo and Pérez Caldentey, 2019). The international dimension of redistribution remains largely unexamined. Yet for any country that imports a significant share of its food, energy, or capital goods, the fiscal cost of a transfer is only part of the story. The transfer generates demand, part of which falls on imports. Imports require foreign exchange. Foreign exchange requires export earnings, reserves, or borrowing — and borrowing capacity depends on the country’s position in the international monetary hierarchy. A transfer that looks affordable in domestic-budget terms can become self-defeating when its import content encounters a binding settlement constraint.

The institutional form of the transfer compounds this problem. Cash and in-kind provisioning are not interchangeable even when they achieve the same welfare gain at the moment of introduction. A cash transfer enters the recipient’s budget and is spent according to the recipient’s preferences, which in an import-dependent economy means a substantial fraction leaks into traded-goods demand and, through that channel, into foreign-exchange requirements. An in-kind housing provision — represented in this paper by a community land trust (CLT) that supplies housing as a physical asset rather

than a monetary entitlement — removes land rent from the transfer cost, reduces the import content of the provisioning, and preserves its real value regardless of domestic price movements. The recipient is housed whether the currency depreciates or not. This is not an argument that in-kind provision is universally superior: it is confined to housing, it does not address food, and it carries costs in quality, location, and spatial concentration that the model does not measure. But the macrofinancial footprint of the two modalities differs, and that difference interacts with the settlement constraint in ways that existing models do not capture.

No published model, to our knowledge, places these elements together. The UBI literature lacks a closed multi-country world in which one country’s transfer spending changes another country’s import prices. The currency-dominance literature derives the reserve-currency advantage from trade-finance complementarities (Gopinath and Stein, 2021) or safe-asset demand (Farhi and Maggiori, 2018), but does not trace the advantage through to social-transfer feasibility. The automation-and-redistribution literature recognises that AI may deteriorate the terms of trade for developing countries (Korinek and Stiglitz, 2021) but treats the adjustment as a relative-price channel, not a hard settlement constraint. Two recent unpublished studies — a New Keynesian small open-economy model of scaled cash transfers (Mendes, Miyamoto, Nguyen, and Pennings, 2026) and an agent-based stock-flow-consistent model of post-automation UBI (Wossnig, 2026) — occupy adjacent territory but differ in structure: neither closes the world, neither maps modality viability as a function of structural position, and neither derives monetary-hierarchy asymmetry from settlement mechanics.

This paper constructs a multi-bloc dynamic computational model that fills this gap. Twenty-five economic blocs, positioned on a two-dimensional plane defined by creditworthiness and food self-sufficiency, trade in a closed world where every import is another bloc’s export and every capital outflow is another’s inflow. Households allocate income and transfers across food, general consumption, and housing through a Stone–Geary demand system. Four redistributive policy regimes — universal cash, targeted cash, food vouchers, and in-kind housing — are calibrated to deliver the same welfare at introduction. A foreign-exchange settlement constraint binds non-reserve blocs: they can import only what they can pay for with export earnings, reserves, and borrowing. A post-employment transition, driven by a single automation parameter, simultaneously raises productive capacity, lowers unit costs, reduces labour income, and shifts income toward capital — forcing the question of whether transfers can fill the gap before the settlement constraint closes it.

The model produces five principal findings. First, welfare-equivalent policy regimes generate sharply different macrofinancial footprints: in-kind housing requires less fiscal outlay and less import demand than cash for the same welfare gain, not because leakage was assigned a lower coefficient but because the demand composition and price-transmission channels differ structurally. Second, in blocs with low creditworthiness and low self-sufficiency, universal cash transfers make recipients worse off than no transfer at all: the exchange-rate depreciation and import-price increase caused by the transfer exceed the transfer’s value. Third, the feasibility of each modality depends on the bloc’s position on the creditworthiness–self-sufficiency plane, producing a phase diagram in which the set of viable redistributive insti-

tutions varies by structural position. Fourth, post-employment transition amplifies this selection pressure: as automation rises, the feasible set shrinks and the modalities separate. Fifth, cost-of-living adjustment of transfers creates a world-price externality: each bloc’s decision to index its transfers to domestic prices raises traded-goods demand, which raises import prices for all blocs — including the adjusting bloc itself — so that the individually rational choice to protect recipients’ purchasing power collectively undermines it.

The paper contributes to the literature in five specific ways. It provides, to our knowledge, the first closed-world model of redistribution under foreign-exchange settlement constraints, filling a gap between the domestic-fiscal UBI literature and the open-economy currency-hierarchy literature. It maps the viability of transfer modalities as a function of structural position — creditworthiness and self-sufficiency — rather than treating feasibility as a single national threshold. It derives monetary-hierarchy asymmetry from settlement mechanics rather than from parametric privilege, showing that the reserve-currency bloc’s protection is conditional and disappears as the transfer level rises. It demonstrates a world-price externality of cost-of-living-adjusted transfers that has not, to our knowledge, previously been shown in simulation. And it connects the post-employment transition to the institutional selection pressure that the international monetary system imposes on redistribution.

The remainder of the paper is organised as follows. Section 2 reviews the relevant literature across three domains: open-economy constraints on redistribution, transfer modality and provisioning, and post-employment re-

distribution. Section 3 presents the model as a complete system. Section 4 reports the results in five stages. Section 5 examines robustness. Section 6 discusses the implications, including the direction of an exit through locally optimised intensive agriculture. Section 7 states limitations. Section 8 concludes.

2. Literature and Conceptual Framework

2.1. Redistribution under open-economy constraints

The macroeconomic constraints on redistribution in an open economy have been studied extensively, but almost never in connection with the design of social transfers. The Mundell–Fleming framework established that a small open economy cannot simultaneously maintain a fixed exchange rate, free capital mobility, and an independent monetary policy — the so-called impossible trinity. This result has become the textbook framing for international macroeconomic policy, yet the redistribution literature has largely proceeded as though the constraint does not exist. The standard UBI model operates within a single economy, asks whether tax revenue can cover the transfer, and evaluates the effects on labour supply and the income distribution, without modelling the exchange rate, the current account, or the foreign-exchange position that determines whether the country can actually pay for the imports its transfer recipients will demand.

The balance-of-payments constraint on growth, formalised by Thirlwall (1979) and extended to include capital flows by Thirlwall and Hussain (1982), provides a more relevant starting point. In this framework, a country’s long-run growth rate is bounded by the ratio of its export growth to the income

elasticity of its imports. Any fiscal expansion that raises demand — including a large-scale cash transfer — increases imports and tightens the external constraint unless exports grow commensurately. The two-gap models of Chenery and Strout (1966), which distinguished a savings gap from a foreign-exchange gap, are the classical ancestor of the settlement constraint used in this paper: when the binding constraint is foreign exchange rather than domestic savings, the country cannot spend its way to higher output regardless of its fiscal capacity.

The structural inability of many countries to borrow in their own currency — Eichengreen and Hausmann’s (1999) “original sin” — compounds this problem. Eichengreen, Hausmann, and Panizza (2005) showed that original sin correlates with country size rather than with institutional quality or fiscal discipline, which implies that the constraint is structural and not easily removed by policy reform. Countries that cannot borrow externally in their own currency face a currency mismatch: depreciation raises the domestic-currency cost of servicing foreign debt, creating a feedback loop between the exchange rate and the fiscal position. Hausmann and Panizza (2003) found that original sin is robustly associated only with economic size, supporting the treatment of creditworthiness as a structural attribute rather than a policy variable.

The post-Keynesian literature on currency hierarchy places these constraints within a broader framework. De Paula, Fritz, and Prates (2017) argued that the international monetary system is hierarchically ordered, with non-reserve currencies facing structural disadvantages in capital flows, interest-rate setting, and exchange-rate management. Bonizzi, Kaltenbrun-

ner, and Michell (2019) applied this framework to critique modern monetary theory (MMT), arguing that external constraints bind over any plausible policy horizon for developing countries. Prates (2020) proposed that monetary sovereignty itself is a spectrum, determined by the currency’s position in the international hierarchy. These arguments are conceptual allies of the present paper, but they target the job guarantee and MMT, not cash or in-kind transfers, and they do not model the transmission from transfer design to external balance.

A separate literature has made the dominance of the US dollar endogenous. Gopinath and Stein (2021) showed that the invoicing role and the safe-asset role of the dollar are complementary, producing a self-reinforcing equilibrium in which a single currency dominates even when multiple large economies share similar fundamentals. Chahrour and Valchev (2022) derived dollar durability from the need to finance international trade, demonstrating multiple stable steady states. Farhi and Maggiori (2018) located the privilege in the monopoly issuance of safe assets. These models explain why currency dominance persists, but none of them traces the consequence through to the feasibility of different forms of social provisioning. The present paper derives asymmetry from a different mechanism — hard settlement feasibility, defined as the accounting constraint that a country cannot import more than it can pay for — and connects it to the modality of redistribution.

2.2. Transfer modality, provisioning, and decommodification

The macroeconomic literature on cash transfers has produced a robust body of evidence on local price effects, labour supply, and poverty reduction, but it has not addressed the import content or foreign-exchange consequences

of different transfer designs. Egger, Haushofer, Miguel, Niehaus, and Walker (2022) found that a cash transfer amounting to more than fifteen percent of local GDP in rural Kenya produced minimal price inflation and a local multiplier of approximately 2.5. This result is widely cited as evidence that large cash transfers are not inflationary. However, the transfers were financed by an external NGO — effectively a foreign-exchange inflow — so the settlement constraint was pre-financed. The result supports the present paper’s argument rather than contradicting it: the benign outcome depends on a condition that domestically financed national-scale transfers do not meet.

The comparison between cash and in-kind transfers has been studied primarily through the lens of consumption efficiency and targeting. Cunha, De Giorgi, and Jayachandran (2019) found that in Mexican villages, cash transfers produced no detectable price increase while in-kind food transfers lowered prices of the transferred goods by 3.7 percent. The authors noted that price effects were largest where local markets were less integrated with outside markets, demonstrating that transfer modality interacts with market openness. Currie and Gahvari (2008) surveyed the theoretical literature on cash versus in-kind transfers, identifying paternalism, targeting, and political economy as the standard justifications for in-kind provision. The import content of different modalities — the share of spending that generates foreign-exchange demand — has not been systematically studied.

The incidence of demand-side housing subsidies is contested and depends heavily on housing supply elasticity. Susin (2002) estimated that housing vouchers raised rents by sixteen percent on average in the ninety largest US metropolitan areas, producing a net welfare loss for non-recipient low-

income households. Fack (2006) found that in France, seventy-eight cents of each additional euro of housing benefit was captured by landlords through higher rents for new claimants. By contrast, Eriksen and Ross (2015) found no effect on overall US rents, though prices rose for units near the voucher ceiling in cities with inelastic supply. Eerola and Lyytikäinen (2021) found no significant pass-through in Finland. The evidence implies that rent capture is conditional on supply elasticity — precisely the land-constrained urban setting in which the provisioning problem is most acute.

Community land trusts represent one institutional response to this problem. By retaining ownership of the land and leasing it to residents, a CLT removes “the most rapidly appreciating element of housing” from the household’s cost (Bourassa, 2007). Thaden and Rosenberg (2010) documented that CLT foreclosure and delinquency rates remained low through the 2008–2010 US housing crisis, providing evidence that the model preserves value through macroeconomic shocks. However, the CLT literature is almost entirely institutional and domestic. No study examines the macroeconomic, fiscal, or open-economy properties of CLTs. The present paper treats the CLT not as a specific institutional prescription but as a representative case of non-cash, asset-based provisioning — a class of transfers whose real value does not depend on the exchange rate and whose import content differs structurally from that of cash.

The claim that in-kind provision is cheaper than cash is not straightforward. Olsen (2003) showed that US public housing production programmes cost substantially more per unit of housing service than vouchers. The advantage claimed in this paper is therefore not construction cost but two other

properties: the exclusion of land rent from the ongoing transfer cost, and the insulation from settlement constraints that follows from reduced import content. These are macrofinancial properties, not microeconomic efficiency claims.

The classical transfer problem, debated by Keynes (1929) and Ohlin (1929), asked whether an international income transfer worsens the paying country’s terms of trade — a “secondary burden” beyond the transfer itself. Chichilnisky (1980) extended the framework to a North–South model with basic goods, contrasting commodity transfers with income transfers. The present paper reframes the transfer problem for domestic social transfers whose spending leaks abroad through imports. In an import-dependent country, a cash transfer to domestic residents functions as an implicit international transfer to the countries that supply traded goods. The terms “fugitive” and “non-fugitive,” introduced here, distinguish transfer modalities by this property: a fugitive transfer is one whose value can be dissipated through exchange-rate depreciation and import-price increases before it reaches the recipient in real terms; a non-fugitive transfer is one whose real value is embodied in a physical asset that is not exposed to these channels. No prior literature uses this terminology.

2.3. Post-employment redistribution and the missing international dimension

The literature on automation, labour displacement, and redistribution has grown rapidly but remains predominantly domestic in scope. Acemoglu and Restrepo (2018) modelled the race between displacement and new task creation in a single economy. Autor (2015) documented the historical capacity of labour markets to generate new occupations, though recent work acknowl-

edges that AI may displace cognitive tasks at a pace that exceeds historical precedent. Sachs and Kotlikoff (2012) showed that in a closed overlapping-generations model, smart machines can produce “immiserating growth” in which output rises while wages and welfare fall. None of these models addresses the international transmission of displacement or the foreign-exchange consequences of the redistribution it necessitates.

Korinek and Stiglitz (2021) provided the most direct treatment of the international dimension. They argued that AI and automation may deteriorate the terms of trade for developing countries by devaluing the sources of their comparative advantage — labour and natural resources. Their analysis implies that domestic policy measures may be insufficient to provide compensation when the terms-of-trade shift is large enough. However, the mechanism operates through relative prices, not through hard settlement constraints, and the analysis does not model the choice between transfer modalities. Alonso, Berg, Kothari, Papageorgiou, and Rehman (2022) constructed a two-country model in which robot productivity gains favour high-wage countries and capital flows “uphill,” but included no redistribution or settlement mechanism.

The gap this paper addresses lies at the intersection of three literatures that have not been connected. The post-employment literature recognises that labour-income replacement is a structural necessity. The currency-hierarchy literature recognises that external constraints bind for most countries. The transfer-modality literature recognises that the form of a transfer affects its economic consequences. No existing model places all three together. Mendes, Miyamoto, Nguyen, and Pennings (2026) come closest

with a New Keynesian small open-economy model of scaled cash transfers in which real exchange-rate appreciation crowds out exports — but their model is a single economy facing an exogenous rest of the world, with adjustment through relative prices rather than settlement rationing. Wossnig (2026) constructs an agent-based stock-flow-consistent model of post-automation UBI with a current-account leak, but the external sector is parametric and does not derive hierarchy from settlement position. Gahvari and Karimi (2016), studying Iran’s 2011 universal cash-transfer reform under an export constraint, may represent the closest empirical precedent for a settlement constraint interacting with a national transfer programme, but the analysis is country-specific and does not compare modalities. The present paper fills this gap with a closed multi-country computational model that asks how the institutional form of redistribution determines its macrofinancial feasibility under international monetary hierarchy, and how simultaneous labour displacement changes the answer.

3. Model

This section presents the model as a complete system. The development proceeded through iterative construction and debugging over eight phases, but the model described here is the final, internally consistent version. Appendix A reports the verification of the accounting identities. Appendix B collects the equations not shown in the main text, the per-period order of computation (Algorithm B.1), and the parameter values (Tables B.1–B.3).

Blocs are indexed by i , quarters by t , and goods by $k \in \{F, G, H\}$ (food, general goods, housing). The period length is $\Delta = 0.25$ years and the horizon

is 120 quarters (30 years). Rates such as inflation π_{it} , the policy rate r_{it} , and output growth $\hat{g}_{it}$ are annual; wherever they enter a quarterly update they are multiplied by Δ .

3.1. Architecture and scope

The model is a discrete-time, quarterly-step dynamic simulation of a closed world economy consisting of multiple economic blocs that trade goods, exchange currencies, and lend to one another. Each period, the following sequence operates:

Labour income and transfers enter household budgets. Households allocate spending across food, general consumption, and housing through a Stone–Geary demand system with subsistence minima. The resulting demand has an import component that depends on the bloc’s self-sufficiency. Import demand from all blocs aggregates into a world traded-goods market, where excess demand relative to supply capacity generates a demand-pressure index. This index feeds into each bloc’s import prices through its exchange rate. The bloc’s foreign-exchange position — export earnings, reserves, and borrowing capacity — determines whether it can pay for the imports its households demand. If it cannot, imports are rationed. Rationing raises domestic prices in the following period and lowers real provisioning. The resulting household consumption feeds back into the next period’s income, saving, and demand. (Figure 1 provides a schematic; Algorithm B.1 gives the order of computation.)

Automation enters from above: a single automation path reduces labour income and shifts it toward capital, while simultaneously raising potential productive capacity and lowering unit costs. The model thus has two parallel

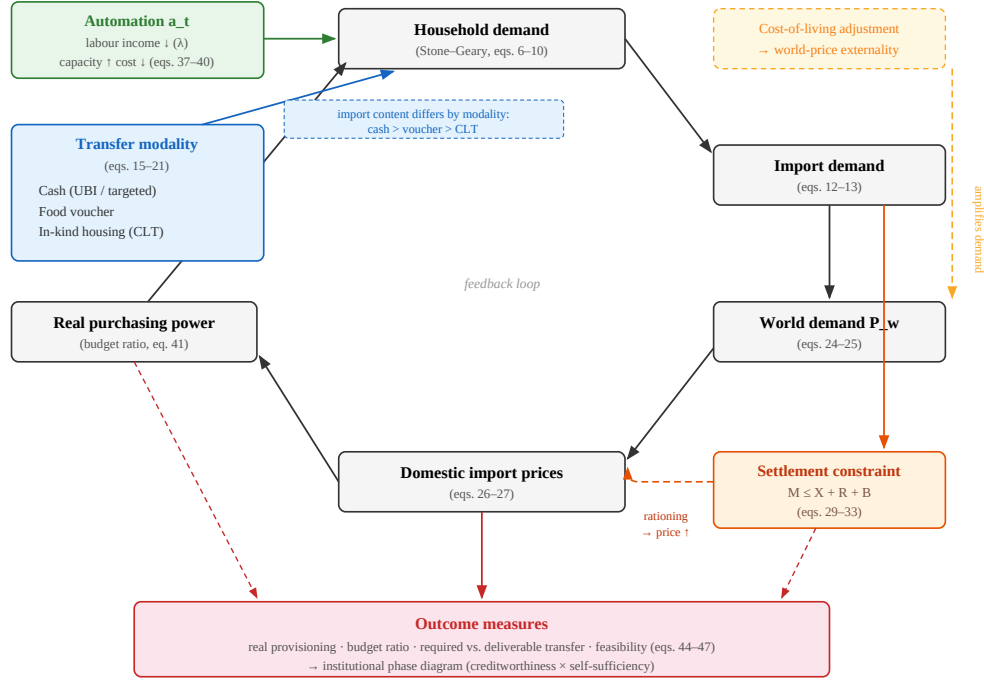


Figure 1: Model architecture. Solid arrows trace the central feedback loop: household demand generates import demand, which feeds world demand pressure, which raises domestic import prices, which reduces real purchasing power, which feeds back to household demand. The settlement constraint (orange) rations imports when foreign-exchange resources are insufficient, raising prices further. Transfer modality (blue) enters through household demand; the import content of spending differs by modality. Automation (green) reduces labour income and raises capacity simultaneously. Cost-of-living adjustment (dashed yellow) amplifies world demand pressure through a cumulative externality. Outcome measures (red) aggregate the loop into feasibility criteria and the institutional phase diagram.

series — a baseline without automation and a post-employment transition with automation at various levels — that share the same structure and differ only in the ceiling of this path.

Three closed-world accounting identities hold at every period: the sum of all current accounts equals zero, the sum of all capital accounts equals zero, and the sum of all changes in net foreign assets equals zero. Writing X_{it} for export earnings, M_{it} for imports, $CA_{it} = X_{it} - M_{it}$ for the current account, K_{it} for the net capital flow, and NFA_{it} for net foreign assets,

$$\sum_i CA_{it} = 0, \quad \sum_i K_{it} = 0, \quad \sum_i \Delta NFA_{it} = \sum_i (CA_{it} + K_{it}) = 0. \quad (1)$$

The identities hold to a tolerance of 10^{-10} (Appendix A).¹ No spending leaks out of the system. A bloc’s import is another bloc’s export revenue. A bloc’s capital outflow is another bloc’s inflow. This closure is a deliberate methodological choice: it ensures that the advantage or disadvantage of any transfer modality cannot arise from spending that disappears at the border.

3.2. *Economic blocs and international accounting*

The world is divided into blocs positioned on a two-dimensional plane. The horizontal axis is creditworthiness: a bloc’s capacity to borrow in international markets, determined by its credit standing, debt ratio, and risk premium. The vertical axis is self-sufficiency: the domestic content of what the

¹The model tracks foreign-exchange reserves R_{it} (equation 33) rather than a full balance sheet of gross foreign assets and liabilities. The closure identity (1) holds on net flows; what it verifies is that the system is closed — no spending leaks out — not that each bloc’s gross asset and liability positions are individually reconciled.

bloc consumes, with food as its defining component. Four archetypes anchor the corners of this plane: a reserve-currency bloc with high creditworthiness and high self-sufficiency, an advanced bloc with high creditworthiness and lower self-sufficiency, an emerging bloc with moderate creditworthiness and moderate self-sufficiency, and a developing bloc with low creditworthiness and low self-sufficiency. A grid of twenty-five blocs populates the plane, with creditworthiness and self-sufficiency varying independently across five levels each.

Each bloc occupies a position $(c_i, \psi_i) \in [0, 1]^2$, with $c_i, \psi_i \in \{0, 0.25, 0.5, 0.75, 1\}$, and every structural parameter ϑ attached to an axis is interpolated linearly between that axis's endpoints:

$$\vartheta_i = \vartheta^{\text{lo}} + (\vartheta^{\text{hi}} - \vartheta^{\text{lo}}) \cdot \text{position}_i. \quad (2)$$

Along the creditworthiness axis, from worst to best, the policy rate runs from 0.070 to 0.020, the baseline foreign-asset share of saving from 0.35 to 0.05, credit standing ς_i from 0.15 to 1.6, and opening reserve standing from 0.25 to 1.75; the foreign-currency share of debt is $f_i = 0.85(1 - c_i)$. Along the self-sufficiency axis, the import contents of food, general goods, and construction run from $(m^F, m^G, m^H) = (0.30, 0.35, 0.25)$ to $(0.10, 0.15, 0.05)$. The reserve issuer sits at $(1, 1)$.

This two-dimensional parameterisation replaces the simpler four-bloc structure used in early phases of the model. Its purpose is to separate two mechanisms that prior work has conflated: the inability to borrow (a creditworthiness problem) and the inability to produce what is consumed (a self-sufficiency problem). A bloc with low creditworthiness but high self-

sufficiency faces a different constraint from a bloc with high creditworthiness but low self-sufficiency. The former cannot borrow but does not need to import much; the latter can borrow but accumulates debt as imports grow. The phase diagram that constitutes the paper’s main result depends on this separation.

Each bloc is further described by an initial GDP and debt ratio, trade openness o_i , a trend growth rate, and a constrained household block (Section 3.3); Table B.2 lists the archetype values. A bloc’s exports are the imports of the other blocs, distributed by supplier shares that depend on economic size, openness, and — when the trade elasticity ϵ_T is active — the supplier’s exchange rate:

$$\begin{aligned}
S_{ij} &= \frac{v_j e_{jt}^{\epsilon_T}}{\sum_{k \neq i} v_k e_{kt}^{\epsilon_T}} \quad (j \neq i), \\
S_{ii} &= 0, \\
v_i &= Y_{i0} o_i, \\
X_{jt} &= \sum_i S_{ij} M_{it}.
\end{aligned} \tag{3}$$

Because each row of S sums to one, $\sum_i X_{it} = \sum_i M_{it}$ holds by construction. The world supply capacity grows along a path calibrated to the realised traded-goods demand of the no-transfer baseline (Section 3.5), ensuring that the model’s supply side does not embed an assumption about the secular trend.

3.3. Household demand

Households within each bloc are divided into two groups: constrained households, whose income is at or below the level needed to meet subsistence

requirements, and unconstrained households, whose income exceeds subsistence. The constrained block is a share n_i of the population receiving a share σ_i of output; both are structural parameters of each bloc, fixed for the duration of the simulation. What changes endogenously is the constrained group's real income, which responds to labour income, transfers, prices, and exchange-rate movements. Its income is

$$I_{it}^c = \sigma_i Y_{it} \phi_{it}^{\text{inc}} \lambda_t, \quad (4)$$

where Y_{it} is bloc output, λ_t is the automation labour-income multiplier (Section 3.7), and ϕ_{it}^{inc} is an income-linkage factor: $\phi_{it}^{\text{inc}} = 1$ when constrained income keeps pace with output, Y_{i0}/Y_{it} when it holds its opening real value, and $Y_{i0}/(Y_{it} \Pi_{it})$ when it is fixed in nominal terms, with Π_{it} the bloc's price level. The post-employment series uses the second rule. Within the constrained block, income can be divided into k equal-population quantiles with different income weights and subsistence tilts (equation B.1); the post-employment series uses five.

Only the constrained block's household problem is solved. The unconstrained block enters through the increment of income that reaches it,

$$\Delta I_{it}^u = \text{cash}_{it}^u + \Delta \text{rent}_{it} + D_{it}, \quad (5)$$

where cash_{it}^u is its share of any universal payment together with land payments under the CLT, Δrent_{it} is the change in rent income, and D_{it} is labour income displaced by automation, which accrues to the owners of the machines (equation 38). The unconstrained block spends a fraction $\text{mpc}^u = 0.3$ of this increment. Because its own consumption problem is never solved,

aggregate provisioning and aggregate demand cannot be summed across the two blocks.

Constrained households allocate spending through a linear expenditure system (LES) derived from Stone–Geary preferences. Each household has subsistence requirements for food (measured in physical units), housing (measured in floor area), and general consumption:

$$\bar{x}_{it}^k = \gamma^k I_{it}^c, \quad (\gamma^F, \gamma^G, \gamma^H) = (0.35, 0.05, 0.25). \quad (6)$$

The requirement is a quantity set in proportion to the group’s current income; it does not respond to prices. The basic-needs bundle used to measure affordability (Section 3.8) is defined separately and held fixed in physical terms throughout the run. Spending above subsistence is allocated across the three categories according to marginal budget shares β^k . Let M be cash income available for market purchases and f^k the quantity of good k received in kind (zero unless a voucher or CLT programme is running). Full income and supernumerary income are

$$\tilde{M} = M + \sum_k p^k f^k, \quad \Omega = \tilde{M} - \sum_k p^k \bar{x}^k, \quad (7)$$

and the interior allocation is

$$x^k = \bar{x}^k + \frac{\beta^k \max(\Omega, 0)}{p^k}, \quad (\beta^F, \beta^G, \beta^H) = (0.25, 0.45, 0.30). \quad (8)$$

The key property of this demand system is that the subsistence requirement does not shrink when prices rise, so that price increases tighten the budget rather than reducing consumption proportionally. The model does

not represent an allocation below subsistence, where Stone–Geary utility is undefined; the bound at zero in (8) is an implementation measure, not a behavioural claim. Where the constraint binds, the reported demand quantity x^k may exceed what the household can actually purchase. This is conservative for the paper’s central finding: it overstates real provisioning in precisely the positions where cash transfers perform worst, so the feasibility boundary reported for cash regimes is, if anything, too generous.

When an in-kind grant exceeds what the household would freely choose, that good’s quantity is pinned at the grant and the remaining budget is reallocated. With binding set $\mathcal{J} = \{k : x^k < f^k\}$ and its complement $\mathcal{J}^c$,

$$\begin{aligned} x^k &= f^k \quad (k \in \mathcal{J}), \\ x^k &= \bar{x}^k + \frac{\beta^k}{\sum_{j \in \mathcal{J}^c} \beta^j} \cdot \frac{\max(\Omega', 0)}{p^k} \quad (k \in \mathcal{J}^c), \\ \Omega' &= M + \sum_{k \notin \mathcal{J}} p^k f^k - \sum_{k \in \mathcal{J}^c} p^k \bar{x}^k. \end{aligned} \tag{9}$$

This is what makes an inframarginal in-kind grant equivalent to cash and an extramarginal one not. Market expenditure and utility are

$$b^k = p^k \max(0, x^k - f^k), \quad U = \sum_k \beta^k \ln(x^k - \bar{x}^k), \tag{10}$$

with $U = -10^9$ assigned when any $x^k \leq \bar{x}^k$.

Housing rent is not given by a formula but clears the rental market. Rental supply is anchored so that the no-transfer economy clears at $p^H = 1$, and the rent p_{it}^H solves

$$\frac{b^H(p^H)}{p^H} = S_i^0 (p^H)^{\varepsilon_i \mu}, \quad (11)$$

$$S_i^0 = \bar{x}^H + \beta^H \left(I_i^c - \bar{x}^F - \bar{x}^G - \bar{x}^H \right),$$

where ε_i is the short-run rental supply elasticity and μ a uniform multiplier on it.

The import content of spending differs across categories. Food has an import content m_i^F set by the bloc's position on the self-sufficiency axis, from 0.30 at its low end to 0.10 at its high end. General consumption has a separate, higher import content m_i^G . Housing spending is largely domestic — construction labour and land are non-traded — but building materials carry an import content m_i^H . Imports are carried in two measures that are not interchangeable — spending, which enters the balance of payments, and quantity, which enters the world market:

$$M_{it}^{\text{sp}} = m_i^F b_{it}^F + m_i^G b_{it}^G, \quad M_{it}^{\text{q}} = m_i^F x_{it}^F + m_i^G x_{it}^G. \quad (12)$$

Import demand facing the world is the no-programme level plus the increment the programme causes,

$$M_{it} = M_{it}^{\text{sp},0} + \Delta M_{it}, \quad (13)$$

$$\Delta M_{it} = \left(M_{it}^{\text{sp},1} - M_{it}^{\text{sp},0} \right) + M_{it}^{\text{gov}} + \text{mpc}^u m_i^G \Delta I_{it}^u,$$

where superscripts 0 and 1 denote household demand solved without and with the programme, and M_{it}^{gov} is the import content of government procurement for in-kind programmes (equation B.3). It is the level, not the increment, that must clear the world market; otherwise a world-wide programme would appear as demand against zero capacity. These import con-

tents translate household demand into foreign-exchange requirements, which is the channel through which the transfer modality affects the settlement constraint.

The constrained block does not save. The unconstrained block saves the fraction $1 - \text{mpc}^u$ of the increment that reaches it and places a share ω_{it} of that saving abroad through a behavioural rule: the foreign-asset share responds to the risk premium rp_{it} (Section 3.6) and to recent exchange-rate depreciation,

$$\omega_{it} = \min \left(0.9, \max \left(0, \omega_i^0 + \nu_\omega (rp_{it} + \max(0, \dot{e}_{it}/\Delta)) \right) \right), \quad (14)$$

$$FA_{it} = (1 - \text{mpc}^u) \Delta I_{it}^u \omega_{it} o_i,$$

where ω_i^0 is the bloc's baseline foreign-asset share, $\nu_\omega = 2.0$, and $\dot{e}_{it}$ is last period's change in the exchange rate. Foreign-asset purchases are a capital outflow, not a purchase of goods: they enter the capital account, not world demand for tradables. This is a reduced-form specification, not an intertemporal optimisation. Section 5 demonstrates that the main results are robust to variation in the parameters of this rule. The sensitivity analysis sweeps the capital-inflow-to-demand parameter from zero to one and shows that the gap between modalities widens rather than closes, confirming that the results do not depend on the specific portfolio closure.

3.4. *Transfer modalities and welfare equivalence*

The model implements four redistributive policy regimes, which differ along two dimensions: targeting (universal versus means-tested) and modality (cash versus in-kind).

Each programme has a size u_i , expressed as a share of GDP. Its budget is a share of current real output when no indexation rule is in force, and a nominal amount T_{it} carried across periods under an indexation rule (equation 22):

$$\mathcal{B}_{it} = \begin{cases} u_i Y_{it} & \text{no indexation rule} \\ T_{it} & \text{under an indexation rule.} \end{cases} \quad (15)$$

Universal cash (UBI) pays out the whole budget, split between the constrained and unconstrained blocks by population share. Targeted cash directs the whole budget to constrained households only, so the per-recipient amount is higher:

$$\begin{aligned} \text{UBI:} \quad & T_{it}^{\text{cash},c} = \mathcal{B}_{it} n_i, \quad T_{it}^{\text{cash},u} = \mathcal{B}_{it}(1 - n_i), \\ \text{targeted:} \quad & T_{it}^{\text{cash},c} = \mathcal{B}_{it}, \quad T_{it}^{\text{cash},u} = 0, \end{aligned} \quad F_{it} = \mathcal{B}_{it}, \quad (16)$$

where F_{it} is the programme's fiscal cost. Food vouchers provide a means-tested entitlement that can be spent only on food; unused portions do not convert to cash, which is the extramarginal case of (9). The grant enters demand as a quantity of food,

$$f_{it}^F = \begin{cases} u_i Y_{it} & \text{no indexation rule} \\ \mathcal{B}_{it}/p_{it}^F & \text{under an indexation rule,} \end{cases} \quad F_{it} = p_{it}^F f_{it}^F, \quad (17)$$

so that under an indexation rule what the budget buys falls as food becomes dearer. In-kind housing provides a physical dwelling — represented as a community land trust (CLT) unit — that removes the housing cost of the granted unit from the recipient's budget; housing consumed beyond the

grant is bought at the market rent, by (10). The land beneath the dwelling is held in trust and does not enter the market. The grant enters demand as a quantity of floor area,

$$f_{it}^H = \begin{cases} u_i Y_{it} & \text{no indexation rule} \\ \mathcal{B}_{it} & \text{under an indexation rule.} \end{cases} \quad (18)$$

Unlike the voucher, the CLT quantity is not divided by a price: once built, the units stay built. Equation (18) specifies the per-period grant; the cumulative housing stock available to recipients is limited by a finite build rate (equation B.4).

The comparison between universal cash and targeted cash isolates the effect of targeting. The comparison among targeted cash, food vouchers, and in-kind housing — all means-tested — isolates the effect of transfer modality holding targeting constant. Results that differ between targeted cash and in-kind housing are therefore attributable to modality, not to the breadth of coverage.

The CLT is treated as a representative case of non-fugitive in-kind asset provisioning, not as a faithful reproduction of any specific CLT programme. Its defining properties for the model are three: land enters the transfer cost at an acquisition discount rather than at market rent, it has lower import content than a cash transfer of equivalent welfare value (because housing services are largely non-traded), and its real value does not depreciate with the exchange rate (because the dwelling exists physically regardless of currency movements). The fiscal cost of in-kind housing splits into structure and land:

$$\begin{aligned}
C_{it}^{\text{build}} &= (1 - \ell_i) f_{it}^H (1 + \kappa_s h_i), \\
C_{it}^{\text{land}} &= \ell_i (1 - \delta_L) \frac{r_{it} + 0.02}{r_{i0} + 0.02} f_{it}^H, \\
F_{it} &= C_{it}^{\text{build}} + C_{it}^{\text{land}},
\end{aligned} \tag{19}$$

where ℓ_i is land's share of gross rent, $\delta_L = 0.5$ the acquisition discount, r_{i0} the pre-transfer interest rate, h_i the policy effort to source construction domestically, and κ_s the cost markup that such sourcing carries, local supply being thinner than the import market. The land payment accrues to landowners and enters cash_{it}^u in (5). Because rent is a flow and construction a stock, the rent the trust collects on its own units is credited against the programme's cost after annualisation at the bloc's own cost of capital over the building's life of $n_L = 40$ years:

$$\begin{aligned}
A(r_i, n_L) &= \frac{r_i}{1 - (1 + r_i)^{-n_L}}, \\
R_{it}^{\text{CLT}} &= \mathcal{R}_{it} \cdot \frac{f_{it}^H}{S_i^0 (p_{it}^H)^{\varepsilon_i \mu} + f_{it}^H} \cdot A(r_i, n_L),
\end{aligned} \tag{20}$$

with $A = 1/n_L$ when $r_i = 0$, and $\mathcal{R}_{it}$ the rent flow in the housing market. A high-rate bloc therefore amortises faster, and the annualised fiscal burden is not a one-time expenditure compared against ongoing cash flows. A finite build rate can phase the programme in (equation B.4). The model does not represent housing quality, locational choice, spatial concentration, or governance costs, all of which would affect the real-world viability of such provision. These omissions are stated as limitations.

All four modalities are calibrated to deliver the same level of household welfare at the moment of introduction. This welfare-equivalent calibration is essential to the paper's argument: the comparison is not between a generous

programme and a stingy one but between programmes that start from the same objective and diverge through their macrofinancial dynamics. The calibration takes as its target the constrained household’s Stone–Geary utility (10) under UBI of size u , evaluated at introduction prices (Y_i at its opening value, $e_i = 1$), and finds for each other modality the size that reaches it:

$$\begin{aligned} U_i^* &= U_i(\text{ubi}, u), \\ u_i^{\text{mod}} &= \{u' : U_i(\text{mod}, u') = U_i^*\}, \\ \text{mod} &\in \{\text{targeted cash, voucher, CLT}\}, \end{aligned} \tag{21}$$

solved by bisection on $u' \in [0, 0.5]$. The search returns a bound rather than a solution in two cases — when it converges below the target utility, and when the required size reaches the ceiling of 0.5 of GDP — and whether either occurs in a reported run is a result, not a given. An equal-cost calibration, which instead equates the fiscal cost F_i/Y_i across modalities, is also available.

Each transfer is subject to a policy rule governing its nominal value over time. Under cost-of-living adjustment, the nominal transfer is increased each quarter by the quarterly fraction of the previous period’s annual consumer-price inflation, preserving the transfer’s purchasing power with a one-period lag. Under nominal freeze, the transfer amount is fixed at its initial value and erodes as prices rise. A third rule, used where indexation is not the object of comparison, keeps the transfer at a fixed share of current real output:

$$T_{it} = \begin{cases} T_{i,t-1} (1 + \pi_{i,t-1} \Delta) & \text{cost-of-living adjustment} \\ T_{i,t^{\text{start}}} & \text{nominal freeze} \\ u_i Y_{it} & \text{output-linked,} \end{cases} \quad T_{i,t^{\text{start}}} = u_i Y_{i,t^{\text{start}}}. \quad (22)$$

Indexing to the previous period's inflation reflects the lag in real indexing arrangements and keeps the transfer and the inflation it causes from becoming simultaneous. The transfer's real value relative to its opening value is

$$\text{RV}_{it} = \frac{T_{it}/\Pi_{it}}{T_{i,t^{\text{start}}}/\Pi_{i,t^{\text{start}}}}, \quad (23)$$

with an analogous measure in terms of rent (equation B.5), which shows that a transfer can hold its value in consumption goods while losing it in housing. The first two rules define the horns of a dilemma that the results section documents: adjustment protects recipients but raises fiscal costs and generates a world-price externality; freezing avoids the externality but abandons recipients to inflation.

3.5. World market and price formation

All blocs' import demands aggregate into a single world traded-goods market. The demand-pressure variable, denoted P_w , is an index of the extent to which aggregate traded-goods demand exceeds aggregate supply capacity. It is a model-internal intermediate variable, not a quantity that corresponds to any observable real-world aggregate. Its role is mechanical: it transmits the aggregate demand pressure into each bloc's import price through the

exchange rate. With Z_t the world's traded-goods supply capacity, the gap is excess world demand as a fraction of capacity, and the index evolves in logs with a slow reversion to a fixed anchor $P_{w,0} = 1$:

$$g_t = \frac{\sum_i M_{it}^q}{Z_t} - 1, \quad (24)$$

$$\hat{\pi}_t^w = \kappa_w g_t - \lambda_w \ln \left(\frac{P_{w,t}}{P_{w,0}} \right), \quad (25)$$

$$P_{w,t+1} = \min \left(\bar{P}, \max \left(10^{-6}, P_{w,t} \exp [\min (\hat{\pi}_t^w \Delta, 20)] \right) \right),$$

with $\kappa_w = 0.35$ and $\lambda_w = 0.05$. The reversion is taken in logs because a restoring term written on the level grows without bound and can drive the index negative. The anchor does not drift, so a transitory gap does not shift the level permanently, while a sustained gap still moves it without limit. The cap $\bar{P} = 10^{12}$ is a reporting limit: a configuration that reaches it is one whose indexation loop ran away.

Each bloc's import price is the index converted at its exchange rate e_{it} and marked up by the scarcity s_{it} left from last period's rationing (Section 3.6). Domestic components cost one by normalisation, scaled by the automation unit-cost multiplier c_t (Section 3.7). Food and general-goods prices are weighted averages of the two:

$$\tau_{it} = \frac{e_{it} P_{w,t}}{\max (10^{-6}, 1 - \min (0.95, s_{it}))}, \quad (26)$$

$$p_{it}^F = \left[(1 - m_i^F) c_t + m_i^F \tau_{it} \right] \varphi_{it}, \quad p_{it}^G = (1 - m_i^G) c_t + m_i^G \tau_{it}, \quad (27)$$

where φ_{it} is a food-price shock multiplier equal to one in the main series. Without scarcity and without a food shock, the food price is $p_{it}^F = (1 - m_i^F) c_t + m_i^F e_{it} P_{w,t}$, where m_i^F is the food import content and e_{it} is the nominal exchange rate. A bloc with high self-sufficiency (low m_i^F) is insulated from movements in P_w ; a bloc with low self-sufficiency is exposed. This is the channel through which one bloc's transfer spending — by contributing to aggregate traded-goods demand — raises another bloc's food prices. Rationing reaches what a household eats in the same way, as a price: a bloc rationed to part of its import demand faces a traded-goods price high enough to clear the quantity it can pay for. Bloc inflation responds to the change in import prices, to the domestic demand pull of the programme, and to a central-bank drag (equation B.7).

The supply capacity of the world grows at a rate calibrated to the no-transfer baseline's realised traded-goods demand, so that the model's supply side does not build in inflationary or deflationary bias. Opening capacity is normalised on the first period's no-transfer demand, so that the baseline opens at a zero gap, and capacity growth is weighted by demand:

$$\begin{aligned} Z_0 &= \sum_i M_{i0}^q, \\ \hat{g}_t^Z &= \frac{\sum_i M_{it}^q \max(0.005, g_i^{\text{base}} - \delta_i t)}{\sum_i M_{it}^q}, \end{aligned} \tag{28}$$

where g_i^{base} is the bloc's trend growth and δ_i its annual decline. The path is fixed once from a no-transfer run, iterated to a fixed point with realised demand, and shared by every configuration, so that a programme's demand cannot call forth the supply to meet it. Under the post-employment tran-

sition, automation raises potential capacity, but actual output is demand-constrained: excess capacity drags on the growth rate rather than being produced and left unsold (Section 3.7).

3.6. International settlement and monetary hierarchy

Each non-reserve bloc faces a settlement constraint: its imports in any period cannot exceed the sum of its export earnings X_{it} , its foreign-exchange reserves R_{it} , and the amount B_{it} it can borrow on international markets:

$$\begin{aligned}\mathcal{K}_{it} &= \max(0, X_{it} + R_{it} + B_{it}), \\ M_{it}^{\text{real}} &= \min(M_{it}, \mathcal{K}_{it}).\end{aligned}\tag{29}$$

Borrowing capacity is a function of the bloc's credit standing ς_i , which is reduced by the contagion state χ_{it} , and of the risk premium, which rises with the debt ratio d_{it} , inflation, and foreign-currency exposure:

$$\begin{aligned}\varsigma_{it} &= \varsigma_i \max(0, 1 - \chi_{it}), \\ B_{it} &= \max\left(0, Y_{it} \bar{b} \Delta \varsigma_{it} \max(0, 1 - \nu_b rp_{it})\right),\end{aligned}\tag{30}$$

$$rp_{it} = \kappa_r \max(0, d_{it} - \bar{d}) + \frac{\kappa_r}{2} \max(0, \pi_{it} - 0.06) + \frac{\kappa_r}{2} f_i \max(0, d_{it} - 0.6) + \kappa_c \chi_{it},\tag{31}$$

with $\bar{b} = 0.05$ per year, $\nu_b = 4$, $\kappa_r = 0.15$, $\bar{d} = 0.9$, and $\kappa_c = 0.05$; at a premium of $1/\nu_b$ the borrowing window shuts. Credit standing is kept separate from the opening debt level: giving a low-credit bloc a low opening debt would make its premium small and its borrowing window wide, inverting the axis. When the constraint binds, imports are rationed — the bloc receives

less than its households demand — and in the following period domestic prices rise to reflect the scarcity, through (26):

$$\begin{aligned} \text{Rationed}_{it} &= \max \left(0, M_{it} - M_{it}^{\text{real}} \right), \\ s_{i,t+1} &= \frac{\text{Rationed}_{it}}{M_{it}}, \\ M_{it}^{\text{q,real}} &= M_{it}^{\text{q}} \frac{M_{it}^{\text{real}}}{M_{it}}, \end{aligned} \tag{32}$$

with $s_{i,t+1} = 0$ when $M_{it} = 0$. Rationing is applied before the world re-clears, because one bloc's cut imports remove another bloc's exports. Reserves evolve as

$$R_{i,t+1} = \max \left(0, R_{it} + X_{it} - M_{it}^{\text{real}} + K_{it}\Delta \right). \tag{33}$$

Here K_{it} is the net capital flow as in (1), not the settlement capacity $\mathcal{K}_{it}$ of (29). The exchange rate responds to capital flows and to the demand for foreign exchange — the current account less the unsettled shortfall, so that a bloc rationed because its reserves ran out is not read as strengthening — with a response proportional to the current rate rather than additive (equation B.6).

The reserve-currency bloc settles in money it issues: while it holds that position, its imports are not rationed and $M_{it}^{\text{real}} = M_{it}$. Reserve-currency status is represented as an institutional difference in settlement capacity — the issuing bloc's import payments do not draw down a finite stock of foreign exchange — rather than through ad hoc preferential coefficients applied to risk premia, capital flows, or exchange-rate adjustment. The position lasts only as long as other blocs want to hold the currency. Reserves are held to

settle future imports, so their attraction is what they will buy:

$$\begin{aligned}
\mathcal{D}_t &= \mathcal{D}_0 \left(\frac{1}{P_{w,t}} \right)^{\epsilon_R}, \\
\mathcal{D}_0 &= \sum_{i: \neg \text{reserve}} Y_{i0} o_i, \\
\text{exempt}_{it} &= \text{reserve}_i \wedge \neg \left[\mathcal{D}_t < \bar{\theta}_R \mathcal{D}_0 \right],
\end{aligned} \tag{34}$$

with $\epsilon_R = 0.5$ and $\bar{\theta}_R = 0.7$, both swept rather than chosen. Conditional advantages then emerge endogenously from the resulting trade, price, and contagion dynamics. At low transfer levels, the reserve bloc’s position on the creditworthiness–self-sufficiency plane naturally protects it: its distance from blocs that experience settlement failure is large, and contagion does not reach it. As the transfer level rises toward $u \approx 0.055$, this protection disappears — not because the model withdraws a privilege but because the number and proximity of distressed blocs increase until the reserve bloc is no longer distant from all of them.

Financial contagion operates through a similarity channel rather than through direct financial linkages. When a bloc experiences settlement failure, lenders reassess blocs with similar structural characteristics — a wake-up-call mechanism (Ahnert and Bertsch, 2022). When bloc j defaults, every other bloc’s contagion state is raised according to its structural distance from j , measured by credit standing and food import content, and the shock decays geometrically over $Y_c = 5$ years:

$$\begin{aligned}
\text{dist}(i, j) &= \sqrt{\left(\frac{|\varsigma_i - \varsigma_j|}{1.6}\right)^2 + \left(\frac{|m_i^F - m_j^F|}{0.25}\right)^2}, \\
\chi_{it} &\leftarrow \min\left(1, \chi_{it} + \varkappa e^{-\text{dist}(i, j)/\varrho}\right), \\
\chi_{i, t+1} &= \chi_{it} \left(1 - \frac{\Delta}{Y_c}\right),
\end{aligned} \tag{35}$$

with reach $\varrho = 0.5$ and strength $\varkappa$. Because the grid moves the three import contents together, distance in m^F is monotone in position on the self-sufficiency axis. The contagion state raises the risk premium (31) and cuts effective credit standing (30), so the reassessment raises risk premia and tightens borrowing capacity for blocs that are structurally proximate to the failed bloc. The reserve issuer is not exempted by assertion; a flight-to-reserve channel routes the withdrawn funds to it as a consequence (equation B.9). This produces a contagion pattern that is ordered by structural position rather than by trade volume or financial exposure.

Policy response is modelled as a reduced-form reaction to gaps relative to the world. A bloc whose growth falls short of the size-weighted world rate $\hat{g}_t^W$, or whose real interest rate falls short of the world yardstick $\bar{r}_t^W$, receives an impulse split between a policy-rate cut and a fiscal boost:

$$\begin{aligned}
\text{gap}_{it} &= \max\left(0, \hat{g}_t^W - \hat{g}_{it}\right) + \max\left(0, \bar{r}_t^W - (r_{it} - \pi_{it})\right), \\
\text{cut}_{it} &= \kappa_P \text{gap}_{it} (1 - \varphi_F), \\
\text{boost}_{it} &= \kappa_P \text{gap}_{it} \varphi_F,
\end{aligned} \tag{36}$$

with $\varphi_F = 0.5$. The rule is identical in every bloc and names no target. The adjustment faces a dilemma: easing depreciates the exchange rate and tightens the settlement constraint; tightening slows growth and induces cap-

ital outflow that depletes reserves. The results show that outcomes do not improve monotonically with the strength of the policy response.

3.7. Post-employment transition

The post-employment transition is driven by automation progress a_t , which approaches a ceiling $a_{\max}$ exponentially, so that the early years move fastest and the last of the work is hardest to automate:

$$a_t = a_{\max} (1 - e^{-\nu_a t}), \quad \nu_a = 0.15 \text{ per year.} \quad (37)$$

From this single path, four consequences are derived simultaneously:

$$\begin{aligned} \Lambda_t &= 1 + \alpha_K a_t, & c_t &= \max(0.05, 1 - \alpha_C a_t), \\ \lambda_t &= \max(0, 1 - \alpha_L a_t), & D_{it} &= \sigma_i Y_{it} (1 - \lambda_t), \end{aligned} \quad (38)$$

with $\alpha_K = 1.0$, $\alpha_C = 0.3$, and $\alpha_L = 1.0$. Potential productive capacity rises by the factor Λ_t , unit costs fall by the factor c_t , the constrained block's labour income is multiplied by λ_t in (4) — so that labour's share of income falls as a consequence — and the income that labour loses, D_{it} , shifts to capital, entering the unconstrained block through (5). Note that $D_{it} = \sigma_i Y_{it} (1 - \lambda_t)$ measures the economy-wide shift from labour income to capital income within the constrained sector's output share. It is not identical to the change in the constrained block's received income, which also depends on the income-linkage factor ϕ_{it}^{inc} in (4). These four effects are not independently calibrated — they are joint consequences of the same technological change, which is the defining feature of a post-employment transition as distinct from a sectoral productivity shock.

Potential output carries the bloc's realised trend Y_{it}^{trend} (equation B.8) multiplied by the automation factor, and the shortfall of output below potential drags on the growth rate:

$$Y_{it}^{\text{pot}} = Y_{it}^{\text{trend}} \Lambda_t, \quad (39)$$

$$\text{slack}_{it} = \max \left(0, 1 - \frac{Y_{it}}{Y_{it}^{\text{pot}}} \right),$$

$$\hat{g}_{it} = g_{it}^{\text{eff}} + 0.08 \eta_f \frac{\Delta \text{Dom}_{it}}{Y_{it}} - \alpha_D \text{slack}_{it} + \sum \text{penalties}_{it}, \quad (40)$$

$$Y_{i,t+1} = \max (0.01, Y_{it} (1 + \hat{g}_{it} \Delta)),$$

where $g_{it}^{\text{eff}} = \max(0.005, g_i^{\text{base}} - \delta_i t)$ is trend growth, ΔDom_{it} the change in domestic demand the programme causes (equation B.2), $\eta_f = 1.2$ a fiscal multiplier, and the penalties are reduced-form terms for capital flows, debt, inflation, and exchange-rate movement (equation B.8). The parameter α_D sets how strongly the demand shortfall holds output down and is swept; at $\alpha_D = 0$ output follows trend growth plus the fiscal impulse regardless of whether what is made can be bought. Output is not capped at potential. When automation raises potential capacity but labour income falls, a gap opens between what the economy could produce and what households can buy, and the resulting slack drags on the growth rate through (40). Transfers fill this gap partially, but the transfer itself generates import demand that encounters the settlement constraint. The question the model answers is whether the transfer can fill the gap before the settlement constraint closes it — and for which blocs, under which policy regime, this is possible.

The model runs two parallel series: a baseline with $a_{\text{max}} = 0$, in which no automation occurs, and a post-employment series with a_{max} at 0.2, 0.4,

0.6, or 0.8. The baseline establishes the benchmark against which the post-employment results are evaluated.

3.8. Outcome measures and feasibility criterion

The model reports outcomes in real physical quantities rather than in monetary aggregates. The primary measure is real provisioning: the quantity of housing (floor area) and food (caloric equivalent) that transfer recipients actually secure in each period. This is the only metric that places cash and in-kind transfers on a common footing — a cash transfer that depreciates before it is spent delivers less provisioning than its nominal value suggests.

The budget ratio is defined as the ratio of a constrained household’s total resources (income plus transfer) to the cost of a fixed basic-needs bundle, where the bundle is defined in physical quantities (a specified amount of food, housing, and general consumption at current prices). For income quantile q of the constrained block, with income share ϖ_q and population share n_q ,

$$\begin{aligned} \text{BR}_{igt} &= \frac{I_{igt} + T_{igt}}{\sum_k p_{it}^k \hat{x}_{iq}^k}, \\ I_{igt} &= \sigma_i Y_{it} \phi_{it}^{\text{inc}} \lambda_t \varpi_q, \\ T_{igt} &= T_{it} n_q. \end{aligned} \tag{41}$$

The bundle quantities are fixed once, at a reference income $I_i^{c,\text{ref}}$, and do not move; only prices do:

$$\hat{x}_{iq}^k = \gamma^k I_i^{c,\text{ref}} \theta \zeta_q, \tag{42}$$

where θ scales the floor and ζ_q is the quantile’s subsistence tilt (equation B.1). A budget ratio below one means the household cannot afford the

bundle. The ratio says whether the money reaches the bundle, not whether the household is fed.

The required transfer level, u^* , is the transfer as a share of GDP that closes the largest quantile shortfall relative to that quantile's population share. With a single constrained type this is the whole constrained block; under the five-quantile configuration of the post-employment series it is the quantile with the largest gap, which need not be the poorest by income. Because the ratio is linear in the transfer, the requirement at given prices is solved directly:

$$T_{it}^* = \frac{1}{\max(0.05, RV_{it})} \max_q \left[\frac{\max(0, \sum_k p_{it}^k \hat{x}_{iq}^k - I_{igt})}{n_q} \right], \quad (43)$$

$$u_{it}^* = \frac{T_{it}^*}{Y_{it}}.$$

The divisor RV_{it} from (23) puts the indexation rules on a common footing: a transfer that has lost half its real value must have been twice as large to begin with. The requirement is a time series, not a constant: as the currency falls and the demand-pressure index rises, the bundle costs more, and the blocs where it grows fastest are those least able to pay. For the fixed point below, the requirement is taken as its maximum over the run's sampled years.

Because the transfer itself changes prices, exchange rates, and import costs, u^* is not a simple arithmetic calculation but a fixed point: the transfer level at which the budget ratio equals one, taking into account all the equilibrium adjustments that the transfer induces. The model searches for this fixed point on a discrete candidate grid $\mathcal{U} = \{0, 0.01, 0.02, 0.03, 0.04, 0.06, 0.08, 0.12\}$ for at most six rounds:

$$\begin{aligned}
u^{(0)} &= u_i^*|_{u=0}, \\
u^{(n+1)} &= \min \left\{ u \in \mathcal{U} : u \geq u_i^*|_{u=u^{(n)}} \right\}, \\
u_i^{\text{FP}} &= u^{(n+1)} \text{ if } u_i^*|_{u=u^{(n+1)}} \leq u^{(n+1)}.
\end{aligned} \tag{44}$$

The fixed point is therefore grid-valued, and “no fixed point” means that none was found on this grid within six rounds.

The deliverable transfer level is the largest candidate whose average rationing share over the run stays within a tolerance ε :

$$\begin{aligned}
\bar{s}_i|_u &= \frac{1}{N_s} \sum_t s_{it}|_u, \\
u_i^{\text{del}}(\varepsilon) &= \max \left\{ u \in \mathcal{U} : \bar{s}_i|_u \leq \varepsilon \right\}, \\
\varepsilon &\in \{0, 0.05, 0.10\}.
\end{aligned} \tag{45}$$

When the required level exceeds the deliverable level, the bloc is in a provisioning trap: it cannot deliver what its population needs without exhausting its capacity to import.

Each bloc is classified at each automation level by four conditions:

$$\begin{aligned}
\textbf{already constrained:} & \quad \bar{s}_i|_{u=0} > 10^{-12}, \\
\textbf{no fixed point:} & \quad u_i^{\text{FP}} \text{ undefined} \wedge u_i^*|_{u=0} > 0, \\
\textbf{trapped}(\varepsilon) : & \quad u_i^*|_{u=0} > 0 \wedge \bar{s}_i|_{u=0} \leq 10^{-12} \\
& \quad \wedge \left(u_i^{\text{FP}} \text{ undefined} \vee u_i^{\text{del}}(\varepsilon) \text{ undefined} \vee u_i^{\text{FP}} > u_i^{\text{del}}(\varepsilon) \right), \\
\textbf{feasible:} & \quad \text{none of the above.}
\end{aligned} \tag{46}$$

A bloc is already constrained when its imports are rationed even without any transfer. The four conditions are evaluated independently and are not

mutually exclusive; the classification applies them in order of precedence — already constrained, no fixed point, trapped, feasible. The no-fixed-point state, which emerges at high automation levels, means that every additional unit of transfer raises prices and import costs enough to require more than one additional unit — the system diverges.

Crisis outcomes are reported as continuous excess measures rather than as a single crisis count. For each bloc, the debt ratio, inflation, and the exchange rate are compared with reporting cut-offs of 1.5, 0.15, and 2.0:

$$\begin{aligned} \text{exc}_{it} &= \left(\frac{\max(0, d_{it} - 1.5)}{1.5}, \frac{\max(0, \pi_{it} - 0.15)}{0.15}, \frac{\max(0, e_{it} - 2.0)}{2.0} \right), \\ \text{severity}_i &= \sum_t \left(\sum \text{exc}_{it} \right) \Delta, \\ \text{duration}_i &= \sum_t \mathbb{1}[\text{exc}_{it} \neq 0]. \end{aligned} \tag{47}$$

Onsets are counted on transitions into crisis, and insolvency is recorded when the debt ratio exceeds 3.0, after which the bloc’s state stops being updated. The cut-offs are reporting thresholds, not structural boundaries. Outcomes are thus reported along five dimensions — onset, duration, severity, insolvency year, and breadth (number of blocs simultaneously in crisis) — rather than as a single crisis count. This decomposition addresses a specific criticism of an earlier version of the model. World-level pressure is reported as levels and changes of the demand-pressure index, the world gap, and the share of demand coming from solvent blocs, with no verdict threshold (equation B.12); imports and foreign-asset purchases are reported separately, because only the first buys goods on the world market (equation B.13).

4. Results

Results are presented in five stages, ordered by the logic of the argument rather than by the sequence in which the model was constructed.

4.1. Same welfare, different macroeconomic footprints

When all four policy regimes are calibrated to deliver the same welfare gain to constrained households at the moment of introduction, they produce sharply different macroeconomic consequences over a thirty-year horizon. Table 1 reports the key indicators for a representative low-creditworthiness, low-self-sufficiency bloc (the developing archetype) under the baseline scenario with cost-of-living adjustment.

Three patterns are visible. First, the required fiscal scale differs across modalities even though the initial welfare objective is identical. Universal cash requires the largest outlay because the payment goes to all households; targeted cash is smaller because it reaches only constrained households; in-kind housing is smaller still because it removes the housing component from the monetary transfer and replaces it with a physical asset whose cost does not include land rent. Second, the import demand generated by each modality differs because the spending composition differs. Cash transfers are spent according to household preferences, which in a low-self-sufficiency bloc means a large fraction goes to imported food and goods. Housing provision generates demand primarily for construction labour and materials, of which the import share is lower. Third, and most strikingly, in-kind housing improves food provisioning despite distributing no food. The mechanism is indirect: by removing rent from the household budget, housing provision frees in-

Table 1: Welfare-equivalent comparison at year 30, developing bloc, baseline with cost-of-living adjustment ($u = 0.02$, all blocs introducing simultaneously). Fiscal scale is the outlay at introduction. Import demand is cumulative real imports over the run, indexed to the no-transfer run. Depreciation is the exchange rate at year 30 against its year-1 value. Food and housing are the physical quantities secured by constrained households, each indexed to its own run’s year 1. The budget ratio is the poorest quantile’s income plus transfer over the cost of the fixed subsistence bundle. The settlement constraint is inactive in this configuration; Sections 4.3 and 4.5 report what happens when it binds.

Modality	Fiscal scale (% GDP)	Import demand index	Depre- ciation (%)	Food (phys.)	Housing (phys.)	Budget ratio (Q1)
UBI	2.0	93.1	+234	149.7	136.7	1.710
Targeted cash	1.0	99.8	+128	180.9	190.1	2.230
Food voucher	1.0	101.1	+156	170.8	175.6	2.016
In-kind housing	0.4	101.5	+89	188.7	198.9	2.254
<i>No transfer</i>	0.0	100.0	+77	191.3	197.4	2.184

come for food spending. At year thirty, the developing bloc’s constrained households secure housing quantities approximately equal to the no-transfer baseline (198.9 versus 197.4 on the year-1 index), while under universal cash the same households secure only 136.7 — less than they would have had with no transfer at all. The same ordering holds for food (188.7 for in-kind housing, 149.7 for universal cash, against 191.3 with no transfer).

This last result deserves emphasis. It means that in the developing archetype, a universal cash transfer calibrated to improve welfare at introduction makes recipients materially worse off over a thirty-year horizon than doing nothing. The transfer’s macrofinancial consequences — exchange-rate depreciation and import-price increases caused by the aggregate demand it generates — take away more than the transfer gives. This is not a theoretical curiosity; it is a transfer reversal — a state in which the transfer itself becomes the proximate cause of welfare deterioration. It differs from the provisioning trap defined in Section 3.8, where the required transfer level exists but exceeds what the settlement constraint can deliver. Here the transfer is delivered, but the price it sets in motion undoes its benefit.

4.2. Transmission mechanism decomposition

The divergence between modalities operates through a chain of transmission that can be decomposed into distinct steps. The chain runs as follows: the transfer modality determines the fiscal scale; the fiscal scale and spending composition determine sectoral demand; sectoral demand determines import demand; import demand contributes to world traded-goods demand pressure; the demand-pressure index feeds through exchange rates into each bloc’s import prices; import prices determine the real purchasing power of the trans-

fer; and real purchasing power determines the provisioning that households actually secure.

At each step, the modalities diverge further. Universal cash generates the highest fiscal outlay, the largest exchange-rate depreciation (+234 percent over thirty years, against +77 percent with no transfer at all), the largest import-price increase, and therefore the smallest real provisioning. In-kind housing generates the smallest fiscal outlay in monetary terms, the least depreciation (+89 percent), and therefore the largest real provisioning.

The import column of Table 1 does not follow that ordering, and the reason is worth stating because it is easy to misread. Universal cash generates the *lowest* cumulative real imports of the four (93.1 against the no-transfer run's 100.0), below in-kind housing's 101.5, while depreciating the currency three times as far. This is not because universal cash asks less of the world market: its marginal import leak is in fact the largest of the four modalities (0.40 against in-kind housing's 0.19, in the units of Appendix B.3). It is because the depreciation it sets off makes imports expensive enough that the realised quantity ends up below what the same bloc would have bought with no programme at all. The level of imports and the pressure that produced it are therefore not the same measurement, and it is the price, not the quantity, that decides what households secure.

Food vouchers, despite being means-tested, concentrate all spending into a single category — food — so their import demand per unit of welfare can exceed that of universal cash, even though general goods carry a higher import content per unit of spending ($m_i^G > m_i^F$). This is why vouchers produce more fixed-point failures than universal cash at automation 0.4 (Section

4.4): the constraint is not the total fiscal outlay but the import composition of spending.

A second distinction runs through the capital account rather than the current account. Universal cash is the only regime that pays the unconstrained block (equation 16): the share $(1 - n_i)$ of the budget that reaches unconstrained households enters the saving channel and, through the foreign-asset allocation rule (equation 14), generates capital outflows that deplete reserves and depreciate the exchange rate. Targeted cash, vouchers, and in-kind housing direct the full budget to constrained households, who do not save in the model. The capital-flight channel is therefore UBI-specific. Vouchers and targeted cash cannot leak through the capital account at all — but vouchers concentrate their entire demand on imported food, while UBI spreads demand across categories and loses part of its budget to capital outflow rather than goods demand. The net effect is that UBI’s capital-account leak, paradoxically, reduces its current-account pressure relative to a voucher of equivalent welfare: the money that flees the country does not buy imported food.

There is a second, independent channel that operates only for housing provision. When a CLT supplies housing directly, the market rent that constrained households would otherwise have paid is removed from the demand side of the housing market. This does not raise the supply of housing in the market (the CLT units are additional), but it does reduce the competition for market-rate housing among unconstrained households, exerting modest downward pressure on market rents. The freed budget — income that would have gone to rent — is redirected to food and general consumption. This is

why housing provision improves food outcomes: it is not a food programme, but it loosens the budget constraint in the direction of food.

The decomposition also explains why universal cash makes recipients in the developing archetype worse off than no transfer. The transfer adds to the household's income, but that income is spent in an economy with high import dependence. The currency depreciates by 234 percent over thirty years, against 77 percent in the same bloc with no transfer at all, and the resulting rise in import prices affects all households, including the transfer recipients themselves. When the indirect induced penalties — all channels removed from the model except exchange-rate and import-price effects — are eliminated, seventy-nine to eighty-five percent of the shortfall against the no-transfer baseline remains. The damage is not driven by behavioural penalties or stigma effects but by the macrofinancial arithmetic of settlement.

4.3. Monetary hierarchy and heterogeneous feasibility

The twenty-five-bloc plane reveals that different structural positions produce different kinds of failure, and that the type of failure — not just its occurrence — depends on the two axes independently.

Insolvency is primarily structured by creditworthiness. Blocs with low creditworthiness exhaust their borrowing capacity first, because they begin with lower reserves, face higher risk premia, and have smaller stocks of net foreign assets to draw down. A bloc at creditworthiness level one reaches settlement failure earlier and at lower transfer levels than a bloc at level three, holding self-sufficiency constant. This is the channel through which original sin operates in the model: the inability to borrow is the proximate cause of import rationing.

Import compression is primarily structured by self-sufficiency. Blocs with low self-sufficiency need more imports for the same level of provisioning, so they draw down reserves faster and hit the settlement constraint sooner. But a surprising interaction emerges: blocs with high self-sufficiency are partially protected even at low creditworthiness. Because they import less, their current-account deficits are smaller, debt accumulates more slowly, and the settlement constraint binds later or not at all. At self-sufficiency 0.892, a bloc at the lowest creditworthiness level avoids insolvency entirely at moderate transfer levels, whereas the same creditworthiness at self-sufficiency 0.689 produces insolvency within fifteen years.

The reserve-currency bloc's protection is conditional. At low transfer levels ($u = 0.04$), it receives a contagion exposure of 0.477 while all other blocs receive 1.000 — a protection that arises purely from its distance on the two-axis plane from the blocs that experience failure. At $u \approx 0.055$, this protection disappears: enough blocs have failed, and the surviving blocs are close enough to the reserve bloc on at least one axis, that the wake-up-call contagion reaches it. The boundary is not a crisis count but a geometric property of the distribution of failures on the plane.

Policy response offers no escape for blocs at structurally disadvantaged positions. Monetary easing depreciates the exchange rate, which tightens the settlement constraint by raising the domestic-currency cost of imports. Monetary tightening slows growth, which induces capital outflows that deplete reserves. The result is non-monotonic: outcomes for blocs with low creditworthiness and low self-sufficiency do not improve continuously with the strength of the policy response.

4.4. *Post-employment transition shrinks the feasible set*

Under the baseline scenario (no automation), provisioning traps appear only when three conditions are met simultaneously: no compression of consumption below subsistence is tolerated, the basic-needs floor exceeds seventy-five percent of the reference level, and the transfer is not adjusted for inflation. Remove any one condition and the trap disappears.

When automation is introduced, all three conditions become unnecessary. At automation level 0.4, traps appear even when moderate consumption compression is tolerated. At 0.6, traps appear even at a basic-needs floor of fifty percent. At 0.4 and above, traps appear whether or not the transfer is adjusted for inflation — though nominal freeze produces more trapped positions (nineteen) than cost-of-living adjustment (eight) at automation 0.4.

The required transfer level rises steeply with automation. At automation 0.8, the required transfer to maintain the poorest quintile’s budget ratio at one reaches approximately 9.6 percent of GDP. The deliverable transfer — the maximum that can be sustained without triggering settlement failure — is approximately 8.0 percent of GDP under a tolerance of ten percent import rationing. Because the required level exceeds the deliverable level, these blocs are in a provisioning trap: the fixed point u^* exists on the candidate grid but lies outside the feasible range. At still higher automation levels, a distinct and more severe state emerges: no candidate on the grid $\mathcal{U}$ achieves $B(u) \geq 1$ within six rounds — every additional unit of transfer raises prices and import costs by more than one unit, and the solver does not converge.

At automation 0.4, the four regimes separate sharply. Of the twenty-five positions, six are already constrained (rationed with no transfer). Among

the remaining nineteen, in-kind housing retains feasibility at all nineteen positions with no fixed-point failures. Targeted cash is feasible at sixteen positions with three fixed-point failures. Universal cash is feasible at twelve with seven failures. Food vouchers are feasible at only nine with ten failures — worse than universal cash, because the voucher’s extramarginal constraint channels all spending into food imports, concentrating import demand in the category with the highest price sensitivity. The ordering of regimes by feasibility is therefore not a simple cash-versus-in-kind ranking: CLT dominates, followed by targeted cash, then universal cash, then vouchers. But even in-kind housing eventually loses feasibility at the highest automation levels for the most structurally disadvantaged positions.

4.5. Cost-of-living adjustment externality and coordination failure

Cost-of-living adjustment of transfers — increasing the nominal payment each period by the previous period’s consumer-price inflation — generates a world-price externality that has not, to our knowledge, previously been shown in simulation.

The mechanism is straightforward. An adjusted transfer tracks prices upward, which maintains the recipient’s purchasing power but also maintains the recipient’s demand for traded goods. That demand contributes to world traded-goods demand pressure, which raises import prices for all blocs, which raises consumer prices, which triggers a further adjustment. The loop does not explode — each iteration is damped by the one-period lag and by the domestic production share — but over thirty years it produces a cumulative difference.

When all twenty-four non-reserve blocs adjust their transfers and the

counterfactual is that none adjust, six to nine of those blocs experience worse budget ratios than they would have had without adjustment. A representative low-creditworthiness, low-self-sufficiency bloc illustrates the mechanism: the real value of its transfer is 0.40 units higher than under nominal freeze (the adjustment works), but its budget ratio is 0.13 lower (the import-price increase caused by everyone’s adjustment more than offsets the gain). The remaining blocs see smaller losses that do not reverse the sign of the budget-ratio change but do reduce the gain from adjustment. The collectively rational decision — refrain from adjustment — is undermined by the individually rational incentive to adjust, because nominal freeze guarantees erosion of the real transfer value.

This is a world-price externality with the structure of a coordination problem. Each bloc’s adjustment maintains its own demand for traded goods, which contributes to world demand pressure, which raises import prices for all blocs. The food export-restriction literature documents an analogous mechanism: each country that restricts exports to insulate its domestic food price raises the world price that all other countries face (Martin and Anderson, 2012). Confirming that non-adjustment is individually unsustainable — that a single bloc freezing its transfer while others adjust is made worse off — would require a one-bloc deviation experiment that is not part of the present design, which models only world-wide simultaneous adoption.

4.6. Institutional feasibility phase diagram

The phase diagram — the paper’s culminating result — displays the interaction between structural position, transfer modality, and automation visually. Two panels show the twenty-five positions on the creditworthiness–

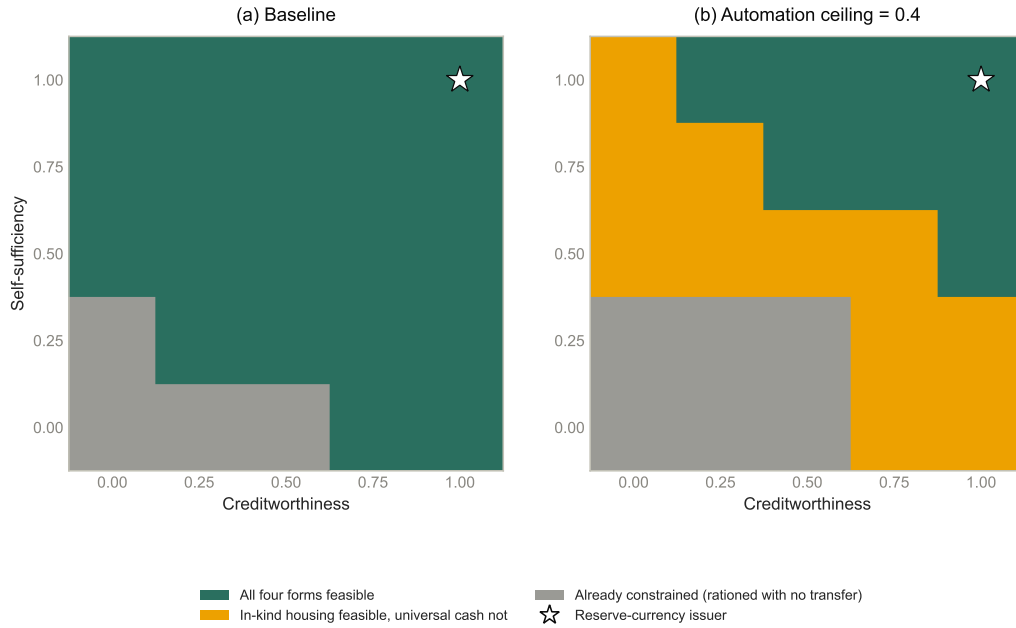


Figure 2: Institutional feasibility by creditworthiness and self-sufficiency. Each cell is one of 25 blocs. Colour shows the set of policy regimes that remain feasible over 30 years. Under automation (b), the feasible frontier shifts inward and the regimes separate. Positions already rationed with no transfer at all are shown separately, because there the constraint is not a statement about what is handed out. A fourth state — no form feasible — is defined but occurs at no position in either panel, and is therefore absent from the legend.

self-sufficiency plane, one for the baseline and one for automation at 0.4. Each position is coloured by the set of policy regimes that remain feasible. Under the baseline, the feasible sets are nearly identical across regimes: all four regimes are feasible at every position that is not already constrained. Under automation 0.4, the sets separate. In-kind housing retains feasibility at every non-constrained position. The cash regimes and the voucher lose feasibility at positions with low creditworthiness or low self-sufficiency, but at different rates and in a pattern that is not simply ordered by the two axes: at $(0.00, 0.75)$, universal cash is feasible but targeted cash shows no fixed point, because the higher per-recipient amount under targeting concentrates demand pressure. The boundary between feasible and infeasible shifts inward — toward higher creditworthiness and higher self-sufficiency — for all regimes except in-kind housing. The figure shows that the international monetary system does not merely limit how much redistribution a country can afford. It selects which institutional forms of redistribution can survive.

5. Robustness

The main results — the divergence of macrofinancial footprints across modalities, the phase-diagram structure, and the shrinkage of feasible sets under automation — are tested against variation in the model’s key assumptions. Full sensitivity tables are reported in Supplementary Material; this section summarises the findings.

Portfolio and capital-flow closure. The foreign-asset allocation rule is behavioural rather than derived from intertemporal optimisation. To assess sensitivity to one key aspect of this specification, the capital-inflow-to-

demand parameter — which governs how much of capital inflows translates into real goods demand — is swept from zero (no pass-through) to one (full pass-through). The gap between policy regimes widens as the parameter increases: at zero, the traded-goods demand-pressure index under universal cash is 1.465 and under in-kind housing is 1.158; at 0.5, the corresponding values are 4.255 and 1.390. The widening occurs because capital inflows under cash transfers amplify domestic demand, which feeds back into imports and exchange-rate pressure. The result confirms that the main findings are robust to alternative assumptions about the real-demand pass-through of capital inflows and, if anything, are conservative under the baseline specification. A full sweep of the portfolio-allocation parameters (ω_0, ω_s) is reported in Supplementary Material.

Cost-of-living adjustment rules. Three rules are compared: full adjustment to previous-period CPI, partial adjustment (fifty percent of CPI growth), and nominal freeze. Full adjustment produces fewer provisioning traps than nominal freeze at all automation levels, but the world-price externality is larger. Partial adjustment lies between the two. The qualitative structure of the phase diagram — modality-dependent feasible sets ordered by structural position — is invariant across rules.

Subsistence floor. The basic-needs floor is varied from fifty to one hundred percent of the reference level in increments of five percentage points. At fifty percent, provisioning traps under the baseline disappear entirely. At seventy-five percent, traps appear only when all three baseline conditions are met. At one hundred percent (no compression tolerated), traps appear more widely. Under automation at 0.4 or above, traps appear at every floor level,

confirming that the post-employment result does not depend on the choice of subsistence threshold.

Contagion strength. The wake-up-call contagion parameter is varied from zero (no contagion) to twice the baseline value. Without contagion, the number of blocs experiencing settlement failure is lower, but the phase-diagram structure persists because the primary driver is each bloc’s own settlement arithmetic. Stronger contagion accelerates failure for structurally proximate blocs and shifts the reserve-currency bloc’s protection boundary to lower transfer levels. The qualitative ordering of modalities is unchanged.

Transfer level. The transfer is swept from zero to twelve percent of GDP. The divergence between modalities is small at low transfer levels and grows monotonically. Below two percent of GDP, all four modalities produce similar outcomes because the macrofinancial footprint is small relative to the economy. Above five percent, the separation becomes pronounced. This confirms that the results are about large-scale redistribution, not about marginal transfers.

World supply elasticity. When the world supply capacity responds more elastically to demand, the demand-pressure effect is dampened and the gap between modalities narrows. When supply is inelastic — as it would be under physical capacity constraints or during a supply-chain disruption — the gap widens. The results are sensitive to this parameter in magnitude but not in sign or structure.

Housing and CLT construction assumptions. The import content of housing construction is varied from ten to forty percent. At higher import content, the advantage of in-kind housing narrows because more of its cost generates

foreign-exchange demand. At forty percent import content, in-kind housing still dominates cash in the developing archetype, but the margin is reduced by approximately half. This confirms that the advantage of housing provision depends in part on the non-traded character of construction, which varies across countries.

6. Discussion

6.1. Redistribution as institutional design

The central finding of this paper is that welfare-equivalent transfers can produce radically different macrofinancial outcomes. This means that the question “how much redistribution can a country afford?” is incomplete without the companion question “in what institutional form?” The fiscal cost of a transfer is one dimension of its feasibility; its import content, its exposure to exchange-rate movements, and its contribution to world demand pressure are others. A country that can afford a cash transfer in domestic-budget terms may be unable to sustain it in settlement terms, while the same welfare objective delivered through in-kind asset provision may remain feasible.

This is not a new insight in the abstract — the humanitarian literature has long recognised that market conditions affect the choice between cash and in-kind aid — but to our knowledge the present paper provides the first formal model in which the institutional form of redistribution interacts with international monetary constraints to determine feasibility. The result generalises beyond the specific modalities modelled here: any transfer whose

spending composition generates high import demand will face tighter settlement constraints than one whose composition is more domestically oriented, holding the welfare objective constant.

6.2. Monetary hierarchy as institutional selection

The phase diagram shows that the international monetary system does not merely limit the quantity of redistribution. It selects which institutional forms of redistribution can survive at each structural position. A country with low creditworthiness and low food self-sufficiency faces a smaller set of viable redistributive institutions than a country with high creditworthiness and high self-sufficiency. The reserve-currency country faces the largest set, but even its feasible set contracts as the transfer level rises.

This selection pressure is structural, not deliberate. No agent in the model designs the monetary hierarchy to favour one form of redistribution over another. The selection arises from the arithmetic of settlement: different policy regimes generate different import demands, which encounter different settlement capacities, which produce different real outcomes.

The world-price externality documented in Section 4.5 adds a collective-action dimension to this selection. When all countries simultaneously adjust their transfers for inflation, the aggregate effect raises world traded-goods demand and import prices for all countries, including the adjusting countries themselves. Six to nine of twenty-four non-reserve blocs end up with worse budget ratios than they would have had without adjustment; the remaining blocs see their gains from adjustment reduced. Whether non-adjustment is individually sustainable — whether a single country that freezes its transfer while others adjust is made worse off — remains untested in the present

design, which models only world-wide simultaneous adoption (Section 4.5). The externality is structurally analogous to the food export-restriction externality documented by Martin and Anderson (2012), in which each country’s price-insulation policy raises the world price that all other countries face.

6.3. Post-employment amplification

Under the baseline scenario, the selection pressure is visible only at the margins of the creditworthiness–self-sufficiency plane. Most positions can sustain all four regimes. The post-employment transition amplifies the selection pressure by simultaneously raising the required transfer level (because labour income falls) and tightening the settlement constraint (because the transfer generates more import demand than the shrinking labour income can finance through exports). At automation 0.4, the feasible sets diverge: in-kind housing retains feasibility at every non-constrained position, while universal cash and food vouchers lose it at seven and ten positions respectively. At higher automation levels, even in-kind housing loses feasibility at the most structurally disadvantaged positions.

This amplification justifies the paper’s title and framing. The post-employment transition is not a speculative scenario appended to the model for completeness; it is the condition under which the model’s central findings become practically relevant. Under the baseline, the institutional form of redistribution is a design choice with moderate macrofinancial consequences. Under post-employment, it becomes a binding constraint that determines whether provisioning is possible at all.

6.4. *The exit direction: self-sufficiency is not fixed*

The model treats food self-sufficiency as structural over the simulation horizon, but it is a policy variable over longer horizons. The provisioning trap depends on food import dependence, and food import dependence can be moved. Historical evidence demonstrates that locally optimised, ecologically embedded intensive agriculture can achieve high per-unit productivity while absorbing large quantities of labour: Mesoamerican chinampa agriculture produced thirteen times the output of dry-land farming, and Andean terrace systems sustained millions without market exchange (Sophia, 2026). The inverse farm-size–yield relationship, established since Sen (1962, 1966) and robust below approximately three to five hectares (Muyanga and Jayne, 2019), provides the modern empirical foundation. The strongest counterargument — Foster and Rosenzweig’s (2022) estimate that consolidating farms to optimal scale would raise income per worker by sixty-eight percent — assumes that displaced agricultural labour is absorbed by the non-farm sector. Under post-employment conditions, that assumption fails (Chiarella, Meyfroidt, Abeygunawardane, and Conforti, 2023). Investments in food-system resilience and import-substituting productive capacity may therefore shift a bloc upward on the phase diagram, but this transition requires low dependence on imported inputs, as Sri Lanka’s 2022 experience demonstrates (FAO and WFP, 2022). Modelling that transition is left for future work.

7. Limitations

The model is a computational exploration, not a calibrated forecasting tool. Several limitations constrain the interpretation of its results.

Household portfolio allocation is represented by a behavioural rule rather than derived from intertemporal optimisation. The sensitivity analysis in Section 5 shows that the main results are robust to variation in this rule, but the model cannot speak to the dynamics of portfolio rebalancing under uncertainty.

The economic blocs are stylised archetypes positioned on a two-dimensional plane, not calibrated representations of specific national economies. The results identify structural patterns — which positions are vulnerable to which failures under which modalities — but do not predict outcomes for any named country.

The model does not include a banking sector. Credit creation, interbank lending, and the amplification of shocks through bank balance sheets are absent. In a world where financial crises often originate in the banking sector, this is a significant omission.

Political adaptation beyond the modelled policy response is not represented. Governments in practice adjust fiscal and monetary policy in complex ways that depend on political constraints, institutional capacity, and public pressure. The reduced-form policy response captures the qualitative dilemma (easing versus tightening) but not the full range of political responses.

Post-collapse dynamics are not modelled. The paper’s motivation — that government failure can lead to civil disorder, armed conflict, and refugee flows — is precisely the set of outcomes that the model does not represent. The

model identifies the economic conditions under which provisioning fails; what follows from that failure is outside its scope.

Within-bloc household heterogeneity is limited to the constrained/unconstrained division and to income quantiles within the constrained block; the unconstrained block's consumption problem is not solved. The model does not represent urban–rural differences, regional variation, or occupational structure within a bloc.

The automation pathway is stylised. Automation progress follows a fixed exponential path toward its ceiling, and its consequences are derived from a single structural relationship. The model does not estimate the speed, sectoral incidence, or upper bound of automation.

Housing quality, locational choice, and spatial concentration are not modelled. In-kind housing provision in the model delivers a homogeneous unit of floor area. In practice, publicly provided housing varies enormously in quality and location, and concentrated provision can produce spatial segregation and stigma. These costs may be substantial and would reduce the advantage of in-kind provision relative to cash.

The basic-needs level is not calibrated to any empirical standard. Results are reported as response surfaces across a range of floor levels, so the reader can assess the sensitivity, but the model does not claim to identify a specific subsistence threshold.

Each bloc's position on the creditworthiness–self-sufficiency plane is fixed for the duration of the simulation. In practice, a bloc that invested in domestic food production or in export capacity could move to a less constrained position over time. Modelling such transitions would require specifying an

investment pathway whose speed, cost, and political feasibility are not predictable from the model’s structure; the phase diagram should therefore be read as showing which positions are vulnerable under current structural conditions, not as a permanent sentence.

The subsistence floor that enters household choice (equation 6) is proportional to current income, whereas the bundle used to measure affordability (equation 42) is fixed in physical quantities. The two are deliberately different — a floor proportional to income cannot fall short, because if income halves so does the floor — but they are not the same object, and results on the budget ratio refer to the fixed bundle.

8. Conclusion

This paper asked whether every country can sustain the provisioning of its population when labour income falls simultaneously across the world economy. The answer is that it depends on what form the provisioning takes and where the country stands in the international monetary hierarchy.

A multi-bloc computational model with closed-world accounting, foreign-exchange settlement constraints, and welfare-equivalent calibration shows that transfer modalities that are identical in their immediate welfare gain produce radically different macrofinancial trajectories. Cash transfers generate high import demand, depreciate the exchange rate, and raise domestic prices through the traded-goods channel; in structurally disadvantaged positions — low creditworthiness, low food self-sufficiency — they make recipients worse off than no transfer at all. In-kind asset provisioning reduces import content,

excludes land rent from the transfer cost, and preserves real value through exchange-rate movements, retaining feasibility at positions where cash has failed. Post-employment transition amplifies this divergence: as automation raises the required transfer level, the feasible set of institutions contracts, and the modalities separate. Cost-of-living adjustment of transfers, individually rational, generates a world-price externality that collectively undermines the provisioning it was designed to protect.

The implication extends beyond the specific modalities and parameters examined here. Open-economy constraints do not merely limit the quantity of redistribution that a country can afford. They shape the institutional forms through which redistribution remains economically feasible. The international monetary system, through the arithmetic of settlement, exerts a selection pressure on redistributive institutions — a pressure that intensifies as the need for redistribution grows. The design of redistribution is therefore not only a question of domestic political economy but an international structural problem whose solution requires attention to the form of provisioning, not only its scale.

Declaration of generative AI and AI-assisted technologies in the writing process

During the preparation of this work the author used Claude (Anthropic) to assist with simulation code development, literature search, equation extraction from code, and manuscript drafting. The author reviewed and edited

all AI-generated content and takes full responsibility for the content of the publication.

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Appendix A. Accounting identities

The model’s central premise is that the world is closed: nothing a bloc spends leaves the system, and nothing a bloc receives arrives from outside it. This appendix reports the verification of that premise. The identities are not approximations imposed on the solution but properties of how the quantities are constructed, and they are checked at every step of every run.

Appendix A.1. The identities

Let M_{it} be bloc i ’s nominal imports in quarter t , X_{it} its exports, $CA_{it} = X_{it} - M_{it}$ its current account, K_{it} its net capital account, and $\Delta NFA_{it} = CA_{it} + K_{it}$ the change in its net foreign asset position. Three identities hold in every quarter of every configuration:

$$\sum_i M_{it} = \sum_j X_{jt}, \quad \sum_i CA_{it} = 0, \quad \sum_i K_{it} = 0. \quad (\text{A.1})$$

The first follows from construction. Imports are distributed across suppliers by a row-stochastic matrix S , with $X_{jt} = \sum_i S_{ij} M_{it}$ and $\sum_j S_{ij} = 1$ for every i , so world exports equal world imports by summation rather than by adjustment. The second is then immediate. The third is imposed: each bloc’s desired capital flow is computed from its own return differential, those desires need not be mutually consistent, and the excess is removed in proportion to bloc size. The fourth identity, $\sum_i \Delta NFA_{it} = 0$, follows from the second and third. The world cannot accumulate claims on itself.

Appendix A.2. Verification

Two checks are run. The first exercises the accounting functions directly on randomly drawn worlds — bloc sizes, openness, import demands and

desired capital flows all sampled independently — and measures the residual. Over 2,000 such worlds the worst absolute residual is 1.8×10^{-15} for trade, 1.2×10^{-15} for the current account, 1.5×10^{-16} for the capital account, and 1.2×10^{-15} for net foreign assets. All are at the floor of double precision. The tolerance asserted in the test suite is 10^{-10} , which is five orders of magnitude looser than anything observed.

The second runs the full simulation and sums the closing net foreign asset positions across blocs. Over a thirty-year horizon with all twenty-five blocs and every policy regime, the sum is -8.9×10^{-17} with no transfer, 6.8×10^{-17} under in-kind housing, and below 10^{-5} under universal cash — the last figure limited by the five-decimal rounding applied when positions are written out, not by the computation.

Appendix A.3. Identities under rationing

The settlement constraint complicates the first identity, because rationing removes imports that another bloc was counting on selling. Cutting bloc i 's imports to what it can pay for therefore removes the corresponding exports from its suppliers. The world is re-cleared on the settled quantities: exports are recomputed from realised rather than desired imports, and the current accounts follow. The identities hold on the quantities actually transacted, which are the ones the balance of payments records.

Reserves move as the residual,

$$R_{i,t+1} = \max \left(0, R_{it} + X_{it} - M_{it}^{\text{realised}} + K_{it}\Delta \right), \quad (\text{A.2})$$

so a bloc cannot settle with foreign exchange it never held. The floor at zero is what makes the constraint bind rather than allowing an implicit overdraft.

Appendix A.4. Nominal and real

One distinction carries the identities. Trade, the capital account and reserves are nominal: they are what the balance of payments is denominated in, and the identities above are stated in those terms. World capacity is a real quantity, so the demand compared against it must be real too, and the quantities bought — not the spending on them — are carried through from the household block for that purpose. Deflating spending by the world price would not recover them, since imported goods carry a domestic price component as well.

Appendix B. Model specification

This appendix collects the equations not shown in Section 3, the order of computation within each period, and the parameter values. Notation follows Section 3.

Appendix B.1. Household block

Income quantiles. The constrained block can be divided into k equal-population quantiles $q = 0, \dots, k - 1$ with income weights and subsistence tilts

$$w_q = 1 + \rho \left(\frac{2q}{k-1} - 1 \right), \quad \zeta_q = \min \left(1.35, \left(\frac{\bar{w}}{w_q} \right)^{0.5} \right), \quad (\text{B.1})$$

where $\rho = 0.6$ is the income spread and $\bar{w}$ the mean weight. The exponent 0.5 damps the tilt: a full inverse tilt drives the poorest quantile below subsistence before any transfer is made, which would make its utility undefined rather than merely low.

Domestic demand. Spending on domestically produced goods is

$$\text{Dom}_{it} = (1 - m_i^F) b_{it}^F + (1 - m_i^G) b_{it}^G. \quad (\text{B.2})$$

Government procurement. For the CLT, construction is sized in real units, so its import content is already a quantity; domestic-sourcing effort h_i reduces the effective import share without changing the structural parameter. For a voucher, the fiscal figure is spending, so the real quantity procured is that figure divided by the food price. The quantity increment of import

demand deflates the unconstrained block's contribution by the general-goods price:

$$\begin{aligned}
\tilde{m}_i^H &= m_i^H (1 - h_i), \\
M_{it}^{\text{gov}} &= \tilde{m}_i^H C_{it}^{\text{build}} \quad (\text{CLT}), \\
M_{it}^{\text{gov}} &= m_i^F F_{it}, \quad M_{it}^{\text{gov,q}} = m_i^F \frac{F_{it}}{p_{it}^F} \quad (\text{voucher}), \\
\Delta M_{it}^q &= (M_{it}^{q,1} - M_{it}^{q,0}) + M_{it}^{\text{gov,q}} + \frac{\text{mpc}^u m_i^G \Delta I_{it}^u}{p_{it}^G}. \tag{B.3}
\end{aligned}$$

CLT build rate. A finite build rate β_{build} , expressed as a share of the existing housing stock H_i^{stock} per year rather than of GDP, phases the programme in:

$$f_{it}^{H,\text{avail}} = \min \left(u_i Y_{it}, \beta_{\text{build}} H_i^{\text{stock}} \max \left(0, t - t_i^{\text{start}} \right) \right). \tag{B.4}$$

Real value of the transfer in housing. Alongside the consumer-price measure (23),

$$\text{HV}_{it} = \frac{T_{it}/p_{it}^H}{T_{i,t^{\text{start}}}/p_{i,t^{\text{start}}}^H}. \tag{B.5}$$

Both measures equal one before the programme starts.

Appendix B.2. Macroeconomic dynamics

Exchange rate. What presses on the rate is the demand for foreign exchange, not the payments actually made, so the unsettled shortfall is added back. The response is proportional to the current rate: an additive rule moves the rate by the same number of points whether it stands at 2.0 or 0.2, so

a bloc under sustained surplus would walk the rate into the floor and stop responding.

$$\begin{aligned}
\text{pressure}_{it} &= CA_{it} - \text{Rationed}_{it}, \\
\text{flow}_{it} &= \left(K_{it} + \eta_b \frac{\text{pressure}_{it}}{Y_{it}} \right) \Delta, \\
\dot{e}_{it} &= -2 \text{flow}_{it} e_{it} + \eta_x (1 - e_{it}) \Delta, \\
e_{i,t+1} &= \max(0.05, e_{it} + \dot{e}_{it}),
\end{aligned} \tag{B.6}$$

with $\eta_b = 0.5$ and expenditure switching $\eta_x = 0.05$. Under the reserve-privilege sensitivity only (off by default), $\dot{e}_{it} \leftarrow \rho_d \dot{e}_{it}$ and $\dot{e}_{it} \leftarrow \max(\dot{e}_{it}, \underline{e}_R - e_{it})$ for the reserve bloc.

Inflation. Inflation is the sum of three pushes on the change in the rate, not a level the rate is set to:

$$\begin{aligned}
\text{world}_{it} &= \frac{1}{\Delta} \left(\frac{P_{w,t+1} \max(0.05, e_{it} + \dot{e}_{it})}{P_{w,t} e_{it}} - 1 \right), \\
\theta_i &= \min \left(0.9, o_i \frac{m_i^F + m_i^G}{2} \right), \\
\text{pull}_{it} &= 0.15 \eta_f \max \left(0, \frac{\Delta \text{Dom}_{it}}{Y_{it}} \right),
\end{aligned}$$

$$\begin{aligned}
\Delta \pi_{it} &= [0.3 \theta_i \text{world}_{it} + \text{pull}_{it} - 0.1 \max(0, \pi_{it} - 0.03)] \Delta, \\
\pi_{i,t+1} &= \max(-0.02, \pi_{it} + \Delta \pi_{it}), \\
\Pi_{i,t+1} &= \max(10^{-6}, \Pi_{it}(1 + \pi_{i,t+1} \Delta)).
\end{aligned} \tag{B.7}$$

Written as a level, the world term would make an appreciating bloc's inflation track the exchange rate down to the floor, which is not what an appreciation does to a price index in which most items are domestic.

Output: growth penalties and trend. The reduced-form penalties in (40), each switchable, and the trend recursion are

$$\begin{aligned}
&\text{capital: } +0.15 \text{ flow}_{it}, \\
&\text{debt: } -0.02 \max(0, d_{it} - 1.2), \\
&\text{inflation: } -0.03 \max(0, \pi_{it} - 0.05), \\
&\text{exchange rate: } -0.01 |\dot{e}_{it}|, \\
&Y_{i,t+1}^{\text{trend}} = \max\left(0.01, Y_{it}^{\text{trend}} \left(1 + g_{it}^{\text{eff}} \Delta\right)\right), \\
&Y_{i,t+1}^{\text{pot}} = Y_{i,t+1}^{\text{trend}} \Lambda_t.
\end{aligned} \tag{B.8}$$

Trend growth belongs to both numerator and denominator of the output–potential ratio, so it is the automation multiplier alone that opens the gap; deriving potential from Y directly would make that ratio identically $1/\Lambda_t$.

Capital account and flight to reserve. Each bloc’s desired net flow responds to its risk-adjusted rate and growth differentials, less household foreign-asset purchases (14); desires are then balanced to sum to zero by size weight v_i from (3):

$$\begin{aligned}
\tilde{K}_{it} &= o_i \nu_K (r_{it} - rp_{it} - \bar{r}_t) + o_i \nu_g (\hat{g}_{it} - \hat{g}_t^W) - FA_{it}, \\
K_{it} &= \tilde{K}_{it} - \left(\sum_j \tilde{K}_{jt} \right) \frac{v_i}{\sum_j v_j}, \\
W_t &= \sum_{i \neq j} \varkappa e^{-\text{dist}(i,j)/\varrho} Y_{it}, \\
\chi_{Rt} &\leftarrow \max\left(0, \chi_{Rt} - \frac{\phi_R W_t}{\max(10^{-9}, Y_{Rt})}\right), \\
R_{Rt} &+= \phi_R W_t,
\end{aligned} \tag{B.9}$$

with $\nu_K = 0.8$, $\nu_g = 0.4$, and R indexing the reserve bloc. Money withdrawn from a region buys the safest claim on offer, so the issuer receives an inflow because others are being withdrawn from.

Policy rate. With the rate cut from (36),

$$\begin{aligned} r_{it}^{\text{tgt}} &= \max(0, 0.02 + 0.5(\pi_{it} - 0.02) + 0.3rp_{it} - \text{cut}_{it}), \\ r_{i,t+1} &= \max\left(0, \min\left(0.30, r_{it} + 0.2\left(r_{it}^{\text{tgt}} - r_{it}\right)\Delta\right)\right). \end{aligned} \quad (\text{B.10})$$

Capacity overhang.

$$\text{CO}_{it} = \min\left(1, \max\left(0, \frac{Y_{it}}{Y_{it}^{\text{pot}}}\right)\right). \quad (\text{B.11})$$

This is not a utilisation rate for the economy: the unconstrained block's consumption is never solved, so its demand is not in the numerator in any full sense.

Appendix B.3. World pressure and leakage measures

World pressure is reported as levels and changes only; nothing is compared against a cut-off:

$$\begin{aligned} \pi_t^w &= \frac{P_{w,t}}{P_{w,t-1}} - 1 \quad (\text{year on year}), \\ \text{gap}_t &= \frac{\sum_i M_{it}^q}{Z_t} - 1, \\ \text{solvent demand share} &= \frac{\sum_{i:\text{alive}} M_{iT}^q}{\sum_i M_{iT}^q}. \end{aligned} \quad (\text{B.12})$$

The last is reported alongside π^w because π^w can be low for opposite reasons — a world that stayed calm, and a world that went quiet after its blocs defaulted.

Imports and foreign-asset purchases are reported apart:

$$\begin{aligned}
\text{import}_i^{\text{abs}} &= 100 \frac{\sum_t \Delta M_{it}}{\sum_t Y_{it}}, \\
\text{FA}_i^{\text{abs}} &= 100 \frac{\sum_t FA_{it}}{\sum_t Y_{it}}, \\
\text{import per fiscal}_i &= \frac{\sum_t \Delta M_{it}}{\sum_t F_{it}}, \\
\text{FA per fiscal}_i &= \frac{\sum_t FA_{it}}{\sum_t F_{it}}.
\end{aligned} \tag{B.13}$$

Appendix B.4. Order of computation

Algorithm B.1. One run of 120 quarters ($\Delta = 0.25$, 30 years).

Initialisation (once).

1. Solve the shared world capacity path from a no-transfer run, iterating capacity and realised demand to a fixed point (at most 12 rounds, tolerance 10^{-9}).
2. Initialise each bloc: Y_{i0} and d_{i0} from the bloc parameters, $e_{i0} = 1$, $\pi_{i0} = 0.02$, $\Pi_{i0} = 1$, $p_{i0}^H = 1$, $R_{i0} = 0.15 Y_{i0} \times \text{reserve standing}$, $s_{i0} = 0$, $\chi_{i0} = 0$.
3. Fix the trade share matrix S and the weights v .
4. Fix each quantile's subsistence bundle in physical quantities (42).
5. Record opening reserve-currency demand $\mathcal{D}_0$.

Each quarter.

6. **Demands at current prices** (no state written). For each bloc: advance the transfer's nominal amount under the indexation rule using last period's inflation; compute the income factor and the automation

multipliers; solve household demand with and without the programme at prices carrying last period's scarcity; form import demand in spending and quantity terms, domestic demand, and fiscal cost.

7. **Set world capacity** from the shared path (at the first step, normalise on the first period's no-transfer demand).
8. **Clear the world.** Recompute trade shares if the trade elasticity is active; distribute imports across suppliers to obtain exports; form current accounts.
9. **Capital account.** Compute risk premia (including the contagion term) and desired flows; balance them to sum to zero by size.
10. **Settlement.** Compute reserve-currency demand and the issuer's exemption; compute borrowing capacity; ration imports to exports plus reserves plus borrowing. Re-clear exports and current accounts on the settled quantities, update reserves, and store the rationing share for next period's prices.
11. **Record** (every fourth step): bloc series, real quantities secured, budget positions and the required transfer, capacity overhang, and world index, capacity, and demand.
12. **Accrue net foreign assets** for every bloc, insolvent ones included, so that the adding-up identity holds.
13. **Update each solvent bloc's state**, in order: debt; exchange rate; inflation; growth; policy impulse; then write $d, K, e, p^H, \pi, \Pi, Y^{\text{trend}}, Y^{\text{pot}}, Y, r$.
14. **Crisis flags and insolvency** against the reporting cut-offs.
15. **Contagion.** Decay existing shocks; for each default this step, raise every other bloc's contagion state by distance; route withdrawn funds

to the reserve issuer if the flight channel is active.

16. **Advance the demand-pressure index** from the world gap and the reversion term.

All blocs compute demands before any state is written, so that one bloc's imports become another's export revenue within the same quarter rather than leaving the model.

Appendix B.5. Parameters

Table B.1. Global parameters (baseline values).

Symbol	Description	Value
η_f	fiscal multiplier	1.2
ν_K	capital-flow sensitivity to the rate differential	0.8
κ_r	risk-premium coefficient	0.15
—	inflation's compression of the real debt ratio	0.3
—	share of the transfer financed by deficit	0.6
η_b	external position $\rightarrow$ exchange-rate pressure	0.5
η_x	expenditure switching on depreciation	0.05

Symbol	Description	Value
ϵ_T	trade elasticity to the exchange rate	0.0
ν_ω	foreign-asset share's sensitivity to risk	2.0
mpc^u	unconstrained households' marginal propensity to consume	0.3
δ_L	CLT land acquisition discount	0.5
—	debt ratio at which the bloc stops being updated	3.0
μ	multiplier on rental supply elasticity	1.0
κ_s	cost markup for domestic CLT sourcing	0.0
$\gamma^F, \gamma^G, \gamma^H$	subsistence coefficients	0.35, 0.05, 0.25
$\beta^F, \beta^G, \beta^H$	marginal budget shares	0.25, 0.45, 0.30
k	income quantiles in the constrained block	1
ρ	income spread within the block	0.6

Symbol	Description	Value
β_{build}	CLT construction per year, share of housing stock	none
n_L	building life for amortisation (years)	40
κ_w	pass-through from world gap to $\hat{\pi}^w$	0.35
λ_w	reversion of P_w toward its anchor	0.05
—	settlement constraint active	off
—	opening reserves, share of GDP	0.15
$\bar{b}$	new foreign borrowing per year, share of GDP	0.05
ν_b	sharpness of borrowing-window closure	4.0
ϵ_R	reserve demand's elasticity to purchasing power	0.5

Symbol	Description	Value
$\bar{\theta}_R$	share of opening reserve demand below which the exemption lapses	0.7
$\varkappa$	contagion strength	0.0
ϱ	contagion reach	0.5
Y_c	contagion unwinding period (years)	5.0
ϕ_R	flight to reserve on default	0.0
κ_c	contagion premium for a fully hit bloc	0.05
κ_P	policy response strength	0.0
ν_g	capital flow sensitivity to the growth gap	0.4
φ_F	fiscal share of the policy impulse	0.5
$a_{\max}$	automation ceiling	0.0
ν_a	automation speed (per year)	0.15
α_K	potential output gained per unit of automation	1.0

Symbol	Description	Value
α_C	unit cost saved per unit of automation	0.3
α_D	strength of demand-shortfall drag on output	0.0
α_L	labour income lost per unit of automation	1.0
θ	multiplier on the measured subsistence bundle	1.0
—	share of a capital inflow spent on tradables on arrival	0.0
$\rho_d, \underline{e}_R$	reserve-privilege sensitivity: exchange-rate damping and floor	0.5, 0.3

Reporting cut-offs: debt ratio 1.5, inflation 0.15, exchange rate 2.0.

Table B.2. Bloc parameters of the four archetypes.

Parameter	B			
	A (reserve)	(non-reserve advanced)	C (emerging)	D (developing)
Initial GDP Y_{i0}	1.2	0.9	0.5	0.2
Initial debt ratio d_{i0}	0.85	0.70	0.40	0.35
Policy rate r_{i0}	0.020	0.025	0.050	0.070
Trend growth g_i^{base}	0.018	0.020	0.035	0.038
Trend decline δ_i	0.0003	0.0003	0.0005	0.0008
Openness o_i	1.0	1.0	0.75	0.50
Foreign- currency debt share f_i	0.0	0.15	0.45	0.65
Initial capital stock	1.3	1.0	0.45	0.20
Constrained income share σ_i	0.15	0.15	0.20	0.25
Constrained population share n_i	0.30	0.30	0.40	0.50

Parameter	B			
	A (reserve)	(non-reserve advanced)	C (emerging)	D (developing)
Food import content m_i^F	0.10	0.15	0.20	0.30
General- goods import content m_i^G	0.15	0.25	0.30	0.35
Construction import content m_i^H	0.05	0.08	0.15	0.25
Rental supply elasticity ε_i	0.3	0.3	0.4	0.5
Land share of gross rent ℓ_i	0.45	0.40	0.35	0.30
Baseline foreign-asset share ω_i^0	0.05	0.15	0.25	0.35

On the 5×5 grid, the axis-linked parameters are instead interpolated by (2).

Table B.3. Post-employment sweep. Automation ceiling $a_{\max} \in \{0, 0.2, 0.4, 0.6, 0.8\}$; floor share $\in \{0.65, 0.75, 0.85, 0.95\}$; indexation $\in \{\text{cost-of-living adjustment,}$

nominal freeze}; rationing tolerance $\varepsilon \in \{0, 0.05, 0.10\}$; candidate grid $\mathcal{U}$ as in Section 3.8. Run configuration: five income quantiles, CLT capacity covering the whole constrained block, build rate 0.02, settlement active, trade elasticity 1.0, policy response 1.0, contagion strength 1.0, flight to reserve 0.5, constrained income holding its opening real value, $5 \times 5 = 25$ blocs.